



Q1 Which of the statements given below is/are correct?

It is always important and useful to include an 'alt' attribute on 'img' tag in HTML because
A. users who cannot see the image due to vision impairment can have a textual description of the image (which can be spoken aloud by a screen reader).

B. If the image fails to load (slow connection, broken path, etc.) then alt text is displayed instead.

C. SEO (Search Engine Optimization) benefits.

Choose the correct answer from the options given below:

- (A) A only (B) B and C only
(C) C only (D) A, B, and C

Q2 What does the following function $f()$ in 'C' return?

```
int f( unsigned int N ){
    unsigned int counter = 0;
    while (N > 0){
        counter += N & 1;
        N = N >> 1;
    }
    return counter = 1; }
```

- (A) 1 if N is odd, otherwise 0
(B) 1 if N is a power of 2, otherwise 0
(C) 1 if the binary representation of N is all 1's, otherwise 0
(D) 1 if the binary representation of N has any 1's, otherwise 0

Q3 Consider the following recursive function $F()$ in Java that takes an integer value and returns a string value:

```
public static String F(int N){
    if (N <= 0) return " - ";
    return F(N - 3) + N + F(N - 2)
    + N;
}
```

The value of $F(5)$ is:

- (A) $-2 - 25 - 3 - 135$ (B) $-2 - 25 - 1 - 3 - 135$
(C) $-1 - 145 - 2 - 245$ (D) $-2 - 25 - 3 - 1 - 135$

Q4

List I	List II
(Programming Terms)	(Meaning)
A. JNDI	I. Runtime support for running java program
B. RMI	II. The API in support of naming and directory services.
C. JDK	III. The methods provided by the Java development kit and runtime support for calling remote methods.
D. JRE	IV. The compiler and class libraries to develop Java applications.

Choose the correct answer from the options given below:

- (A) A - II, B - III, C - IV, D - I
(B) A - II, B - IV, C - III, D - I
(C) A - I, B - III, C - IV, D - II
(D) A - III, B - II, C - IV, D - I

Q5 Match List I with List II

List I	List II
(Programming Paradigms)	(Characteristics)
A. Imperative	I. Declarative, clausal representation, theorem proving
B. Object-oriented	II. Side-effect-free, declarative, expression evaluation
C. Logic	III. Imperative, abstract data type
D. Functional	IV. Command-based, procedural

- (A) A - IV, B - III, C - I, D - II
(B) A - III, B - IV, C - I, D - II
(C) A - IV, B - III, C - II, D - I
(D) A - II, B - III, C - I, D - IV

Q6



Suppose you have eight 'black and white' images taken with a 1- *megapixel* camera and one '8-color' image taken by an 8- *megapixel* camera. How much hard disk space in total do you need to store these images on your computer?

- (A) 1 GB
- (B) 4 MB
- (C) 3 MB
- (D) 3 GB

Q7 Which of the statements given below are correct?

The midpoint (or Bresenham) algorithm for rasterizing lines is optimized relative to DDA algorithm in that

- A. it avoids round-off operations.
- B. it is incremental.
- C. it uses only integer arithmetic.
- D. all straight lines can be displayed as straight (exact).

Choose the correct answer from the options given below:

- (A) A and B only
- (B) A and C only
- (C) A, B, C, and D
- (D) A, B, and C only

Q8 What is the transformation matrix M that transforms a square in the xy -plane defined by $(1, 1)^T$, $(-1, 1)^T$, $(-1, -1)^T$ and $(1, -1)^T$ to a parallelogram whose corresponding vertices are $(2, 1)^T$, $(0, 1)^T$, $(-2, -1)^T$ and $(0, -1)^T$?

- (A) $M = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (B) $M = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- (C) $M = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (D) $M = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Q9 Suppose a Bezier curve $P(t)$ is defined by the following four control points in the xy -plane: $P_0 = (-2, 0)$; $P_1 = (-2, 4)$; $P_2 = (2, 4)$; and $P_3 = (2, 0)$. Then which of the following statements are correct?

A. Bezier curve $P(t)$ has degree 3.

B. $P\left(\frac{1}{2}\right) = (0, 3)$.

C. Bezier curve $P(t)$ may extend outside the convex hull of its control points.

Choose the correct answer from the options given below:

- (A) A and B only
- (B) A and C only
- (C) B and C only
- (D) A, B, and C

Q10 Given below are two statements

Statement I: The maximum number of sides that a triangle might have when clipped to a rectangular viewport is 6.

Statement II: In 3D graphics, the perspective transformation is nonlinear in z .

In light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true.
- (B) Both Statement I and Statement II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Q11 In software testing, beta testing is the testing performed by _____.

- (A) potential customers at the developer's location
- (B) potential customers at their own locations
- (C) product developers at the customer's location
- (D) product developers at their own locations

Q12 The V components in MVC are responsible for:

- (A) User interface.
- (B) Security of the system.
- (C) Business logic and domain objects.
- (D) Translating between user interface actions/events and operations on the domain objects.

Q13 In software engineering, what kind of notation do formal methods predominantly use?

- (A) textual (B) diagrammatic
- (C) mathematical (D) computer code

Q14



If every requirement stated in the Software Requirement Specification (SRS) has only one interpretation, then SRS is said to be

- (A) correct
- (B) consistent
- (C) unambiguous
- (D) verifiable

Q15 A system has 99.99% uptime and has a mean-time-between-failure of 1 *day*. How fast does the system has to repair itself in order to reach this availability goal?

- (A) ~9 *Seconds*
- (B) ~10 *Seconds*
- (C) ~11 *Seconds*
- (D) ~12 *Seconds*

Q16 In the context of Software Configuration Management (SCM), what kind of files should be committed to your source control repository?

- A. Code files
- B. Documentation files
- C. Output files
- D. Automatically generated files that are required for your system to be used

Choose the correct answer from the options given below:

- (A) A and B only
- (B) B and C only
- (C) C and D only
- (D) D and A only

Q17 Match List I with List II

List I	List II
Software Process Model	Description
A. Waterfall Model	I. Software can be developed incrementally
B. Evolutionary Model	II. Requirement compromises are inevitable
C. Component-based Software Engineering	III. Explicit recognition of risk

D. Spiral Development	IV. Inflexible partitioning of the project into stages
-----------------------	--

(A) A - IV, B - I, C - III, D - II

(B) A - I, B - IV, C - II, D - III

(C) A - II, B - III, C - I, D - IV

(D) A - IV, B - I, C - II, D - III

Q18 Identify the correct order of the following five levels of Capability Maturity Model (from lower to higher) to measure the maturity of an organisation's software process.

- A. Defined
- B. Optimizing
- C. Initial
- D. Managed
- E. Repeatable

Choose the correct answer from the options given below

- (A) C, A, E, D, B
- (B) C, E, A, B, D
- (C) C, B, D, E, A
- (D) C, E, A, D, B

Q19 Given below are two statements

Statement I: Cleanroom software process model incorporates the statistical quality certification of code increments as they accumulate into a system.

Statement II: Cleanroom software engineering follows the classic analysis, design, code, test, and debug cycle to software development and focussing on defect removal rather than defect prevention.

In light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Q20 Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Software developers donot do



exhaustive software testing in practice.

Reason R : Even for small inputs, exhaustive testing is too computationally intensive (e.g., takes too long) to run all the tests.

In light of the above statements, choose the correct answer from the options given below

- (A) Both A and R are true and R is the correct explanation of A
 (B) Both A and R are true but R is NOT the correct explanation of A
 (C) A is true but R is false
 (D) A is false but R is true

Q21 Which of the following are logically equivalent?

- A. $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$
 B. $(p \rightarrow q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$
 C. $(p \rightarrow q) \rightarrow (r \rightarrow s)$ and $(p \rightarrow r) \rightarrow (q \rightarrow s)$

Choose the correct answer from the options given below:

- (A) A only (B) A and B only
 (C) B and C only (D) A and C only

Q22 Which of these statements about the floor and ceiling functions are correct?

Statement I: $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + (1/2) \rfloor$ for all real number x

Statement II: $\lceil x + y \rceil = \lceil x \rceil + \lceil y \rceil$ for all real numbers x and y

- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is true but Statement II is false
 (D) Statement I is false but Statement II is true

Q23 How many ways are there to assign 5 different jobs to 4 different employees if every employee is assigned at least 1 job

- (A) 1024 (B) 625
 (C) 240 (D) 20

Q24 A company stores products in a warehouse. Storage bins in this warehouse are specified by their aisle, location in the aisle, and shelf. There are 50 aisles, 85 horizontal locations in each aisle, and 5 shelves throughout the warehouse. What is the least number of products the

company can have so that at least two products must be stored in the same bin?

- (A) 251 (B) 426
 (C) 4251 (D) 21251

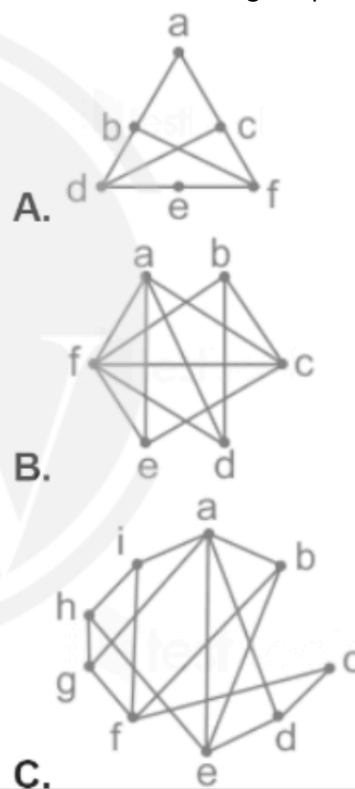
Q25 Let us assume a person climbing the stairs can take one stair or two stairs at a time. How many ways can this person climb a flight of eight stairs?

- (A) 21 (B) 24
 (C) 31 (D) 34

Q26 For which value of n is Wheel graph W_n regular

- (A) 2 (B) 3
 (C) 4 (D) 5

Q27 Which of the following Graphs is(are) planar?



- (A) A and B only (B) B and C only
 (C) A only (D) B only

Q28 Match List I with List II

List I	List II
Identity	Name
A. $x+x=x$	I. Identity Law
B. $x+0=x$	II. Absorption Law
C. $x+1=1$	III. Idempotent Law
D. $x+xy=x$	IV. Domination Law



- (A) A - III, B - I, C - II, D - IV
 (B) A - I, B - III, C - IV, D - II
 (C) A - III, B - I, C - IV, D - II
 (D) A - III, B - IV, C - I, D - II

Q29 Let $(X, *)$ be a semigroup. Furthermore, for every a and b in X , if $a \neq b$, then $a * b \neq b * a$.

Based on the defined semigroup, choose the correct equalities from the options given below:

- A. For every a in X , $a * a = a$
 B. For every a, b in X , $a * b * a = a$
 C. For every a, b, c in X , $a * b * c = a * c$
 (A) A and B only
 (B) A and C only
 (C) B and C only
 (D) A, B and C

Q30 Consider the following linear optimization problem:

$$\begin{aligned} \text{Maximize } Z &= 6x + 5y \\ \text{Subject to } 2x - 3y &\leq 5 \\ x + 3y &\leq 11 \\ 4x + y &\leq 15 \\ \text{and } x &\geq 0, y \geq 0. \end{aligned}$$

The optimal solution of the problem is:

- (A) 15 (B) 25
 (C) 31.72 (D) 41.44

Q31 Which of the following languages are not regular?

- A. $L = \{(01)^n 0^k \mid n \geq k, k \geq 0\}$
 B. $L = \{c^n b^k a^{n+k} \mid n \geq 0, k \geq 0\}$
 C. $L = \{0^n 1^k \mid n \neq k\}$

Choose the correct answer from the options given below:

- (A) A and B only
 (B) A and C only
 (C) B and C only
 (D) A, B and C

Q32 Any string of terminals that can be generated by the following context-free grammar (where S is start nonterminal symbol)

$$S \rightarrow XY$$

$$X \rightarrow 0X \mid 1X \mid 0$$

$$Y \rightarrow Y0 \mid Y1 \mid 0$$

- (A) has at least one 1
 (B) should end with 0
 (C) has no consecutive 0's or 1's
 (D) has at least two 0's

Q33 Let

$$L_1 = \{0^n 1^n 0^m \mid n \geq 1, m \geq 1\}$$

$$L_2 = \{0^n 1^m 0^m \mid n \geq 1, m \geq 1\}$$

$$L_3 = \{0^n 1^n 0^n \mid n \geq 1\}$$

Which of the following are correct statements?

- A. $L_3 = L_1 \cap L_2$
 B. L_1 and L_2 are context free languages but L_3 is not a context free language
 C. L_1 and L_2 are not context free languages but L_3 is a context free language
 D. L_1 is a subset of L_3

Choose the correct answer from the options given below:

- (A) A and B only
 (B) A and C only
 (C) A and D only
 (D) A only

Q34 Given below are two statements

Statement I: The family of context free languages is closed under homomorphism

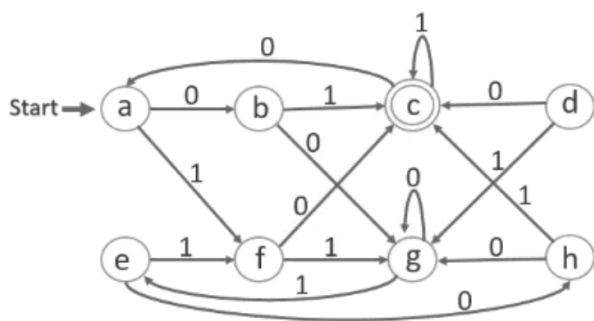
Statement II: The family of context free languages is closed under reversal

In light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is true but Statement II is false
 (D) Statement I is false but Statement II is true

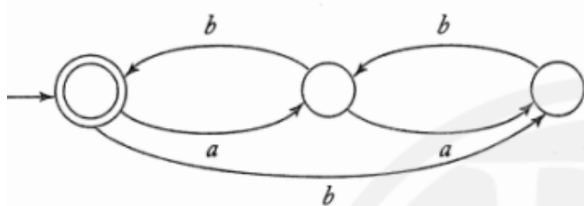
Q35 What is the minimum number of states required to the finite automaton equivalent to the transition diagram given below?





- (A) 3 (B) 4
(C) 5 (D) 6

Q36 Find the regular expression for the language accepted by the automata given below.



- (A) $(aa^*(a+b)ab)^*$
(B) $(a+b)ab(ab+bb+aa^*(a+b)ab)^*$
(C) $a^*ab(ab+bb+aa^*(a+b)ab)^*$
(D) $a^*(a+b)ab(ab+bb+aa^*(a+b)ab)^*$

Q37 What language is accepted by the pushdown automaton

M

=

$(\{q_0, q_1, q_2\}, \{a, b\}, \{a, b, z\}, \delta, q_0, z, \{q_2\})$

with $\delta(q_0, a, a) = \{(q_0, aa)\}$; $\delta(q_0, b, a)$

$= \{(q_0, ba)\}$

$\delta(q_0, a, b) = \{(q_0, ab)\}$; $\delta(q_0, b, b)$

$= \{(q_0, bb)\}$

$\delta(q_0, a, z) = \{(q_0, az)\}$; $\delta(q_0, b, z)$

$= \{(q_0, bz)\}$

$\delta(q_0, \lambda, b) = \{(q_1, b)\}$; $\delta(q_0, \lambda, a)$

$= \{(q_1, a)\}$

$\delta(q_1, a, a) = \{(q_1, \lambda)\}$; $\delta(q_1, b, b)$

$= \{(q_1, \lambda)\}$

$\delta(q_1, \lambda, z) = \{(q_2, z)\}$?

(A) L

$= \{w \mid n_a(w) = n_b(w), w \in \{a, b\}^+\}$

(B) L

$= \{w \mid n_a(w) \leq n_b(w), w \in \{a, b\}^+\}$

(C)

$$L = \{w \mid n_b(w) \leq n_a(w), w \in \{a, b\}^+\}$$

(D) $L = \{ww^R \mid w \in \{a, b\}^+\}$

Q38 Match List I with List II

List I	List II
Production Rules	Grammar
A. $S \rightarrow XY$ $X \rightarrow 0$ $Y \rightarrow 1$	I. Greibach Normal Form
B. $S \rightarrow aS \mid bSS \mid c$	II. Context sensitive Grammar
C. $S \rightarrow AB$ $A \rightarrow 0A \mid 1A \mid 0$ $B \rightarrow 0A$	III. Chomsky Normal Form
D. $S \rightarrow aAbc$ $Ab \rightarrow bA$ $Ac \rightarrow Bbcc$ $bB \rightarrow Bb$ $aB \rightarrow aa \mid aaA$	IV. S. Grammar

Choose the correct answer from the options given below:

- (A) A - III, B - I, C - IV, D - II
(B) A - III, B - II, C - I, D - IV
(C) A - III, B - IV, C - I, D - II
(D) A - IV, B - III, C - I, D - II

Q39 Which of the following concepts can be used to identify loops?

- A. Depth first ordering
B. Dominators
C. Reducible graphs

Choose the correct answer from the options given below:

- (A) A and B only
(B) A and C only
(C) B and C only
(D) A, B and C

Q40 Given below are two statements

Statement I: LL(1) and LR are examples of Bottom-up parsers.

Statement II: Recursive descent parser and SLR are examples of Top-down parsers



In light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Q41 The postfix form of the expression $(A + B) * (C * D - E) * F / G$ is _____

- (A) $A B + C D * E - F G / * *$
- (B) $A B + C D * E - F * * G /$
- (C) $A B + C D * E - * F * G /$
- (D) $A B + C D E * - * F * G /$

Q42 A double-ended queue (deque) supports adding and removing items from both the ends of the queue. The operations supported by deque are AddFront(adding item to front of the queue), AddRear(adding item to the rear of the queue), RemoveFront(removing item from the front of the queue), and RemoveRear(removing item from the rear of the queue). You are given only stacks to implement this data structure. You can implement only push and pop operations. What's the time complexity of performing AddFront() and AddRear() assuming m is the size of the stack and n is the number of elements?

- (A) $O(m)$ and $O(n)$
- (B) $O(1)$ and $O(n)$
- (C) $O(n)$ and $O(1)$
- (D) $O(n)$ and $O(m)$

Q43 Two balanced binary trees are given with m and n elements, respectively. They can be merged into a balanced binary search tree in ____ time.

- (A) $O(m*n)$
- (B) $O(m + n)$
- (C) $O(m*\log n)$
- (D) $O(m*\log(m + n))$

Q44 A data structure is required for storing a set of integers such that each of the following operations can be done in $O(\log n)$ time, where n is the number of elements in the set.

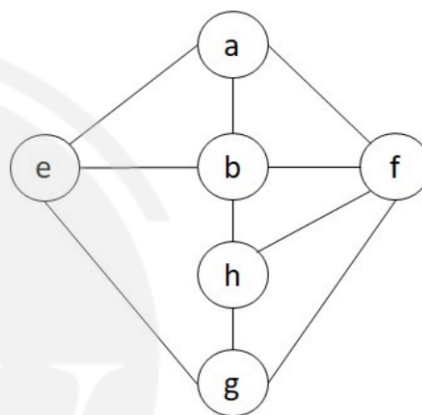
- Deletion of the smallest element

- Insertion of an element if it is not already present in the set

Which of the following data structures can be used for this purpose?

- (A) A heap can be used but not a balanced binary search tree.
- (B) A balanced binary search tree can be used but not a heap.
- (C) Both balanced binary search tree and heap can be used.
- (D) Neither balanced binary search tree nor heap can be used.

Q45 Consider the following graph.



Among the following sequences

- I. a b e g h f
- II. a b f e h g
- III. a b f h g e
- IV. a f g h b e

Which are depth first traversals of the above graph?

- (A) I, II, and IV only
- (B) I and IV only
- (C) II, III, and IV only
- (D) I, III, and IV only

Q46 Given below are two statements

Statement I: In an undirected graph, number of odd degree vertices is even.

Statement II: In an undirected graph, sum of degrees of all vertices is even.

In light of the above statements, choose the correct answer from the options given below.

- (A) Both Statement I and Statement II are true.



- (B) Both Statement I and Statement II are false.
 (C) Statement I is true but Statement II is false.
 (D) Statement I is false but Statement II is true.

Q47 Which of the given options provides the increasing order of asymptotic complexity of functions f_1 , f_2 , f_3 and f_4 ?

- A. $f_1(n) = 2^n$
 B. $f_2(n) = n^{3/2}$
 C. $f_3(n) = n \log n$
 D. $f_4(n) = n^{\log n}$

Choose the correct answer from the options given below

- (A) C, B, D, A
 (B) C, B, A, D
 (C) B, C, A, D
 (D) B, C, D, A

Q48 The order of a leaf node in a $B+$ tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is $1K$ bytes, data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

- (A) 63 (B) 64
 (C) 67 (D) 68

Q49 A hash function h defined as $h(key) = key \bmod 7$, with linear probing, is used to insert the keys $44, 45, 79, 55, 91, 18, 63$ into a table indexed from 0 to 6 . What will be the location of key 18 ?

- (A) 3 (B) 4
 (C) 5 (D) 6

Q50 In the following table, the left column contains the names of standard graph algorithms and the right column contains the time complexities of the algorithms. Here, n and m are number of vertices and edges, respectively. Match each algorithm with its time complexity.

List I	List II
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Standard graph algorithms	Time complexities
A. Bellman-Ford algorithm	I. $O(m \log n)$
B. Kruskal's algorithm	II. $O(n^3)$
C. Floyd-Warshall algorithm	III. $O(n^*m)$
D. Topological sorting	IV. $O(n+m)$

Choose the correct answer from the options given below:

- (A) A - III, B - I, C - II, D - IV
 (B) A - II, B - IV, C - III, D - I
 (C) A - III, B - IV, C - I, D - II
 (D) A - II, B - I, C - III, D - IV

Q51 Which agent deals with the happy and unhappy state?

- (A) Utility-based agent
 (B) Model-based agent
 (C) Goal-based Agent
 (D) Learning Agent

Q52 Given below are two statements

Statement I: Breadth-First Search is optimal when all the step costs are equal whereas uniform-cost search is optimal with any step-cost.

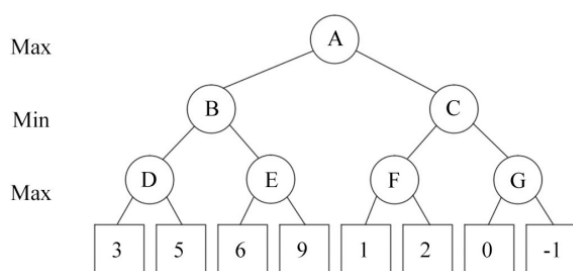
Statement II: When all the step costs are same uniform-cost search expands more nodes at depth d than the Breadth-First Search.

In light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is true but Statement II is false
 (D) Statement I is false but Statement II is true

Q53 Consider the given tree below. Calculate the value at the root of the tree using alpha-beta pruning algorithm.





(A) 3

(B) 5

(C) 6

(D) 9

Q54 Which among the following statement(s) is(are) FALSE?

- A. Greedy best-first search is not optimal but is often efficient.
- B. A^* is complete and optimal provided $h(n)$ is admissible or consistent.
- C. Recursive best-first search is efficient in terms of time complexity but poor in terms of space complexity.
- D. $h(n) = 0$ is an admissible heuristic for the 8-puzzle.

(A) A only

(B) A and D only

(C) C only

(D) C and D only

Q55 Consider the sentence below.

There is a country that borders both India and Pakistan.

Which of the following logical expressions express the above sentence correctly when the predicate $Country(x)$ represents that x is a country and $Borders(x, y)$ represents that the countries x and y share the border?

- (A) $\exists c Country(c) \wedge Border(c, India) \wedge Border(c, Pakistan)$
- (B) $\exists c Country(c) \Rightarrow [Border(c, India) \wedge Border(c, Pakistan)]$
- (C) $[\exists c Country(c)] \Rightarrow [Border(c, India) \wedge Border(c, Pakistan)]$
- (D) $\exists c Border(Country(c), India \wedge Pakistan)$

Directions (56-60) Read the following passage and answer the given questions.

Next five questions are based on the following passage.

Consider a domain consisting of three Boolean variables Toothache, Cavity, and Catch. The full joint distribution is a $2 \times 2 \times 2$ table as shown in the figure below.

	toothache		$\neg$ toothache	
	catch	$\neg$ catch	catch	$\neg$ catch
cavity	0.108	0.012	0.072	0.008
$\neg$ cavity	0.016	0.064	0.144	0.576

Q56 The marginal probability of cavity $P(\text{cavity})$ is _____.

(A) 0.200

(B) 0.216

(C) 0.120

(D) 0.080

Q57 The probability of a cavity, given evidence of a toothache, $P(\text{cavity} \mid \text{toothache})$ is _____.

(A) 0.400

(B) 0.600

(C) 0.280

(D) 0.216

Q58 The probability of a toothache, given evidence of a cavity, $P(\text{toothache} \mid \text{cavity})$ is _____.

(A) 0.400

(B) 0.600

(C) 0.280

(D) 0.216

Q59 $P(\text{cavity} \vee \text{toothache})$ is _____.

(A) 0.200

(B) 0.120

(C) 0.280

(D) 0.600

Q60 The probability for Cavity, given that either Toothache or Catch is true, $P(\text{Cavity} \mid \text{toothache} \vee \text{catch})$ is _____.

(A) 0.6000

(B) 0.5384

(C) 0.8000

(D) 0.4615

Q61 Which of the DBMS component ensures that concurrent execution of multiple operations on the database results into a consistent database state?

(A) Logs

(B) Buffer manager

(C) File manager

(D) Transaction processing system



- Q62** Consider following two statements:
 Statement I: Relational database schema represents the logical design of the database.
 Statement II: Current snapshot of a relation only provides the degree of the relation.
 In the context to the above statements, choose the correct option from the options given below:
 (A) Statement I is TRUE but Statement II is FALSE
 (B) Statement I is FALSE but Statement II is TRUE
 (C) Both Statement I and Statement II are FALSE
 (D) Both Statement I and Statement II are TRUE

- Q63** Given a fixed-length record file that is ordered on the key field. The file needs B disk blocks to store R number of records. Find the average access time needed to access any record of the given file using binary search.

- (A) $\frac{B}{2}$ (B) $B + R$
 (C) B (D) $\log_2 B$

- Q64** Given the following STUDENT-COURSE scheme: STUDENT (Rollno, Name, courseno) COURSE (courseno, coursename, capacity), where Rollno is the primary key of relation STUDENT and courseno is the primary key of relation COURSE. Attribute coursename of COURSE takes unique values only. Which of the following query(ies) will find total number of students enrolled in each course, along with its coursename.

- A. SELECT coursename, count(*) 'total' from STUDENT natural join COURSE group by coursename;
 B. SELECT C.coursename, count(*) 'total' from STUDENT S, COURSE C where S.courseno=C.courseno group by coursename;
 C. SELECT coursename, count(*) 'total' from COURSE C where courseno in (SELECT courseno from STUDENT);
 (A) A and B only
 (B) C only
 (C) A only
 (D) B only

- Q65** Given the following STUDENT-COURSE scheme: STUDENT (Rollno, Name, Courseno) COURSE

(Courseno, Coursename, Capacity), where Rollno is the primary key of relation STUDENT and Courseno is the primary key of relation COURSE. Attribute Coursename of COURSE takes unique values only. The number of records in COURSE and STUDENT tables are 3 and 5 respectively. Following relational algebra query is executed: $R = \text{STUDENT} \times \text{COURSE}$
 Match List I with List II in context to the above problem statement.

List I	List II
A. Degree of table R	I. 15
B. Cardinality of table R	II. NIL
C. Foreign key of relation STUDENT	III. 6
D. Foreign key of relation COURSE	IV. Courseno

Choose the correct answer from the options given below:

- (A) A -III, B -I, C -IV, D - II
 (B) A -I, B -III, C -IV, D - II
 (C) A -I, B -III, C -IV, D - II
 (D) A -I, B -III, C -II, D - IV

- Q66** Suppose a B tree is used for indexing a database file. Consider the following information: size of the search key field = 10 bytes, block size = 1024 bytes, size of the record pointer = 9 bytes, size of the block pointer = 8 bytes. Let K be the order of internal node and L be the order of leaf node of B tree, then $(K, L) =$ _____.
 (A) (57, 53)
 (B) (50, 52)
 (C) (60, 64)
 (D) (34, 31)

- Q67** Given a relation scheme $R(x, y, z, w)$ with functional dependencies set $F = \{x \rightarrow y, z \rightarrow w\}$. All attributes take single and atomic values only.
 A. Relation R is in First Normal FORM
 B. Relation R is in Second Normal FORM
 C. Primary key of R is xz
 Choose the correct answer from the options given below:
 (A) C only (B) B and C only



(C) A and C only (D) B only

Q68 A company is consuming parts in the manufacturing of other products. Each of the part is either manufactured within the company or purchased from the external suppliers or both. For each part, partnumber, partname is maintained. Attribute batchnumber is maintained if the consumed part is manufactured in the company. If part is purchased from external supplier, then supplier name is maintained. Which of the following constraints need to be considered when modelling class/subclass concepts in ERD for the given problem.

- (A) Total specialization and overlapping constraints
(B) Disjoint constraint only
(C) Partial participation
(D) Partial participation and disjoint constraints

Q69 A transaction may be in one of the following states during its execution life cycle in concurrent execution environment.

- A. FAILED
B. TERMINATED
C. PARTIALLY COMMITTED
D. COMMITTED
E. ACTIVE

Given a transaction in active state during its execution, find its next transitioned state from the options given below:

- (A) A only
(B) Either A or C only
(C) C only
(D) D only

Q70 Which of the following is used to create a database schema?

- (A) DML
(B) DDL
(C) HTML
(D) XML

Q71 Given memory access time as p nanoseconds and additional q nanoseconds for handling the page fault. What is the effective memory access

time if a page fault occurs once for every 100 instructions?

- (A) $p + \frac{q}{100}$ (B) $\frac{p+q}{100}$
(C) $p + q$ (D) $\frac{p}{100} + q$

Q72 In a file allocation system, the following allocation schemes are used:

- A. Contiguous
B. Indexed
C. Linked allocation

Which of the allocation scheme(s) given above will not suffer from external fragmentation?

Choose the correct answer from the options given below:

- (A) A only (B) B and C only
(C) A and B only (D) C only

Q73 Given below are three statements related to interrupt handling mechanism

- A. Interrupt handler routine is not stored at a fixed address in the memory.
B. CPU hardware has a dedicated wire called the interrupt request line used for handling interrupts
C. Interrupt vector contains the memory addresses for specialized interrupt handlers.

In the context of above statements, choose the correct answer from the options given below:

- (A) A is TRUE only
(B) Both B and C are TRUE only
(C) Both A and B are TRUE only
(D) Both A, C are TRUE only

Q74 Match List I with List II

List-I	List-II
System Calls	Description
A. fork()	I. sends a signal from one process to particular process
B. exec()	II. Indicates termination of current process
C. kill()	III. Loads the specified program in memory.
D. exit()	IV. Creates a child process.



Choose the correct answer from the options given below:

- (A) A - II, B - III, C - IV, D - I
 (B) A - IV, B - III, C - I, D - II
 (C) A - IV, B - I, C - II, D - III
 (D) A - IV, B - III, C - II, D - I

Q75 Consider the following 3 processes with the length of the CPU burst time given in milliseconds:

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9

What is the average waiting time for these processes if they are scheduled using preemptive shortest job first scheduling algorithm?

- (A) 5.5 (B) 2.66
 (C) 4.66 (D) 6

Directions (76-80) Read the following passage and answer the given questions.

Consider a machine with 16 GB main memory and 32-bits virtual address space, with page size as 4KB. Frame size and page size is same for the given machine.

Q76 The number of bits reserved for the frame offset is:

- (A) 12 (B) 14
 (C) 32 (D) 8

Q77 Find number of pages required for the given virtual address space

- (A) 2^{10} (B) 2^{20}
 (C) 2^{30} (D) 2^{12}

Q78 What is the minimum number of bits needed for the physical address?

- (A) 28 (B) 34
 (C) 24 (D) 12

Q79 What is the size of page table for handling the given virtual address space, given that each page table entry is of size 2 bytes?

- (A) 2MB (B) 2KB
 (C) 32MB (D) 12KB

Q80 If a process of size 34KB is to be executed on this machine, then what will be the size of internal fragmentation for this process?

- (A) 4KB (B) Zero
 (C) 1KB (D) 2KB

Q81 Match List I with List II

List-I	List-II
A. Odd Function	I. NAND Gate
B. Universal Gate	II. XOR Gate
C. 2421 Code	III. Amplification
D. Buffer	IV. Self-complementing

Choose the correct answer from the options given below:

- (A) A - I, B - II, C - III, D - IV
 (B) A - II, B - I, C - IV, D - III
 (C) A - I, B - III, C - II, D - IV
 (D) A - IV, B - II, C - III, D - I

Q82 The Octal equivalent of hexadecimal $(D.C)_{16}$ is:

- (A) $(15.6)_8$ (B) $(61.6)_8$
 (C) $(15.3)_8$ (D) $(61.3)_8$

Q83 A digital computer has a common bus system for 8 registers 16 bits each. How many multiplexers are required to implement common bus? What size of multiplexers is required?

- (A) 16, 8×1
 (B) 8, 16×1
 (C) 8, 8×1
 (D) 16, 16×1

Q84 Which of the following is not an example of pseudo-instruction?

- (A) ORG
 (B) DEC
 (C) END
 (D) HLT

Q85 The reverse Polish notation of the following infix expression

$[A * \{B + C * (D + E)\}] / \{F * (G + H)\}$ is_____.

- (A) $ABCDE + * + *FGH + */$



- (B) $ABCDE * + + *FGH + */$
 (C) $ABCDE + * + *FGH * + /$
 (D) $ABCDE + * * +FGH + */$

Q86 Arrange the following in the increasing order of complexity.

- A. I/O Module
 B. I/O processor
 C. I/O Channel
 D. DMA

Choose the correct answer from the options given below

- (A) D, C, B, A
 (B) C, D, A, B
 (C) A, B, C, D
 (D) A, D, C, B

Q87 The characteristics of the combinational circuits are:

- A. Output at any time is function of inputs at that time
 B. Contains memory elements
 C. Do not have feedback paths
 D. Clock is used to trigger the circuits to obtain outputs

Choose the correct answer from the options given below:

- (A) A and B only
 (B) B and C only
 (C) A and C only
 (D) B and D only

Q88 The cache coherence problem can be solved

- A. by having multiport memory
 B. allow only nonshared data to be stored in cache
 C. using a snoopy cache controller
 D. using memory interleaving

Choose the correct answer from the options given below:

- (A) A and C only
 (B) B and C only
 (C) D and C only
 (D) B and D only

Q89

Given below are two statements

Statement I: CISC computers have a large of number of addressing modes.

Statement II: In RISC machines memory access is limited to load and store instructions.

In light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is true but Statement II is false
 (D) Statement I is false but Statement II is true

Q90 Which of the following statements is true?

- (A) Control memory is part of the hardwired control unit.
 (B) Program control instructions are used to alter the sequential flow of the program.
 (C) The register indirect addressing mode for accessing memory operand is similar to displacement addressing mode.
 (D) CPU utilization is not affected by the introduction of Interrupts.

Q91 Match List I with List II

List I	List II
A. Data Link Layer	I. True end-to-end layer
B. Network Layer	II. Token Management
C. Transport Layer	III. Produce billing information
D. Session Layer	IV. Piggybacking

Choose the correct answer from the options given below:

- (A) A - IV, B - III, C - I, D - II
 (B) A - IV, B - II, C - I, D - III
 (C) A - II, B - III, C - I, D - IV
 (D) A - IV, B - I, C - III, D - II

Q92



Given below are two statements

Statement I: Telnet, Ftp, Http are application layer protocol

Statement II: The Iridium project was planned to launch 66 low orbit satellites.

In light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Q93 Which of the following statement is False?

- (A) Packet switching leads to better utilization of bandwidth resources than circuit switching.
- (B) Packet switching results in less variation in the delay than circuit switching.
- (C) Packet switching sender and receiver can use any bit rate, format or framing method unlike circuit switching.
- (D) Packet switching can lead to reordering unlike circuit switching.

Q94 In Ethernet when Manchester coding is used, the bit rate is _____

- (A) Half the baud rate
- (B) twice the baud rate
- (C) Thrice the baud rate
- (D) Same as the baud rate

Q95 The address of class B host is to be split into subnets with 6-*bit* subnet number. What is the maximum number of the subnets and the maximum number of hosts in each subnet?

- (A) 62 subnets and 262142 hosts
- (B) 64 subnets and 1024 hosts
- (C) 64 subnets and hosts 262142
- (D) 62 subnets and 1022 hosts

Q96 In electronic mail, which of the following protocols allows the transfer of multimedia?

- (A) IMAP
- (B) SMTP
- (C) POP3
- (D) MIME

Q97 A message is encrypted using public key cryptography to send a message from sender to

receiver. Which one of the following statements is True?

- (A) Sender encrypts using receiver's public key
- (B) Sender encrypts using his own public key
- (C) Receiver decrypts using sender's public key
- (D) Receiver decrypts using his own public key

Q98 Which of the following statements are true?

- A. Frequency division multiplexing technique can be handled by digital circuits.
- B. Time division multiplexing technique can be handled by analog circuits
- C. Wavelength division multiplexing technique is used with optical fiber for combining two signals.
- D. Frequency division multiplexing technique can be applied when the bandwidth of a link is greater than the bandwidth of the signals to be transmitted.

Choose the correct answer from the options given below:

- (A) B and D only
- (B) C and D only
- (C) A and D only
- (D) B and D only

Q99 Which of the following statements are true?

- A. X_{25} is connection-oriented network
- B. X_{25} doesn't support switched virtual circuits.
- C. Frame relay service provides acknowledgements.
- D. Frame relay service provides detection of transmission errors.

Choose the correct answer from the options given below:

- (A) A and D only
- (B) B and D only
- (C) C and D only
- (D) B and C only

Q100 Let $G(x)$ be the generator polynomial used for CRC checking. The condition that should be satisfied by the $G(x)$ to catch all errors consisting of an odd number of inverted bits is:

- (A) $(x + 1)$ is factor of $G(x)$
- (B) $(x - 1)$ is factor of $G(x)$



(C) $(x^2 + 1)$ is factor of $G(x)$

(D) $(1 - x^2)$ is factor of $G(x)$



Answer Key

Q1 (D)
Q2 (B)
Q3 (D)
Q4 (A)
Q5 (A)
Q6 (B)
Q7 (D)
Q8 (A)
Q9 (A)
Q10 (D)
Q11 (B)
Q12 (A)
Q13 (C)
Q14 (C)
Q15 (A)
Q16 (A)
Q17 (D)
Q18 (D)
Q19 (C)
Q20 (A)
Q21 (A)
Q22 (C)
Q23 (C)
Q24 (D)
Q25 (D)
Q26 (C)
Q27 (C)
Q28 (C)
Q29 (D)
Q30 (C)
Q31 (D)

Q32 (D)
Q33 (A)
Q34 (A)
Q35 (C)
Q36 (B)
Q37 (D)
Q38 (C)
Q39 (D)
Q40 (B)
Q41 (C)
Q42 (B)
Q43 (B)
Q44 (B)
Q45 (D)
Q46 (A)
Q47 (A)
Q48 (A)
Q49 (C)
Q50 (A)
Q51 (A)
Q52 (A)
Q53 (B)
Q54 (C)
Q55 (A)
Q56 (A)
Q57 (B)
Q58 (B)
Q59 (C)
Q60 (D)
Q61 (D)
Q62 (A)



Q63 (D)
Q64 (A)
Q65 (A)
Q66 (A)
Q67 (C)
Q68 (A)
Q69 (B)
Q70 (B)
Q71 (A)
Q72 (B)
Q73 (B)
Q74 (B)
Q75 (C)
Q76 (A)
Q77 (B)
Q78 (B)
Q79 (A)
Q80 (D)
Q81 (B)

Q82 (A)
Q83 (A)
Q84 (D)
Q85 (A)
Q86 (D)
Q87 (C)
Q88 (B)
Q89 (A)
Q90 (B)
Q91 (A)
Q92 (A)
Q93 (B)
Q94 (A)
Q95 (D)
Q96 (D)
Q97 (A)
Q98 (B)
Q99 (A)
Q100 (A)



Hints & Solutions

Q1 Text Solution:

All three points regarding the use of the alt attribute in the tag in HTML are true:

Accessibility: The alt text provides a description of the image for users with vision impairments, which can be read aloud by screen readers.

Image Fallback: If the image fails to load (due to issues like a slow connection or broken path), the alt text will be displayed instead of the image.

SEO Benefits: The alt attribute helps search engines understand and index the content of images, improving SEO by making the images more discoverable.

Q2 Text Solution:

The correct option is B.

When N=1:

If the function N=1 then,

counter=0,

Iteration 1:

The while-loops enter and the expression

counter += N & 1;

counter = 0+1 & 1;

counter =1

The expression is N = 1 >> 1;

N=0

>> is the right shift operator so N becomes 0 and in the next iteration, the condition becomes false.

The while loop will execute and it returns 1 as output.

Here 1==1 is true hence it prints the 1 as output.

When N=2:

If the function N=2 then,

counter=0,

iteration 1:

The while-loops enter and the expression

counter += N & 1;

counter = 0+2 & 1;

counter =0+0

counter =0

The expression is N = 2 >> 1;

N=1

>> is the right shift operator so N becomes 1 and in the next iteration the condition becomes true.

Iteration 2:

The while-loops enter and the expression

counter += N & 1;

counter = 0+1 & 1;

counter =0+1

counter =1

The expression is N = 1 >> 1;

N=0

>> is the right shift operator so N becomes 0 and the next iteration the condition becomes false.

counter == 1;

Here 1==1 is true hence it prints the 1 as output.

The while loop will execute and it returns 1 as output.

When N=3:

If the function N=3 then,

counter=0,

Iteration 1:

The while-loops enter and the expression

counter += N & 1;

counter = 0 + 3 & 1;

counter =0+1

counter =1

The expression is N = 3 >> 1;

N=1

>> is the right shift operator so N becomes 1 and the next iteration the condition becomes true.

iteration 2:

The while-loops enter and the expression

counter += N & 1;

counter = 1+1 & 1;

counter =1+1

counter =2

The expression is N = 1 >> 1;

N=0

>> is the right shift operator so N becomes 0 and in the next iteration, the condition becomes false.

The while loop will execute and it returns 0 as output.

Here 2==1 is False hence it prints the 0 as output.



Q3 Text Solution:

Option D is Correct

Base Case: The base case is when $N \leq 0$. In this case, the function returns the string "-".

Recursive Case: When $N > 0$, the function makes two recursive calls:

- $F(N-3)$
- $F(N-2)$

The result is: $F(N) = F(N-3) + N + F(N-2) + N$. The recursive calls continue until $N \leq 0$, at which point they return "-".

1. First Call: $F(5)$:

- Compute $F(5-3) = F(2)$
- Compute $F(5-2) = F(3)$
- Result: $F(5) = F(2) + 5 + F(3) + 5$

2. Second Call: $F(2)$:

- Compute $F(2-3) = F(-1) = \text{"-"}$ (base case)
- Compute $F(2-2) = F(0) = \text{"-"}$ (base case)
- Result: $F(2) = \text{"-"} + 2 + \text{"-"} + 2 = \text{"-2-2"}$

3. Third Call: $F(3)$:

- Compute $F(3-3) = F(0) = \text{"-"}$ (base case)
- Compute $F(3-2) = F(1)$
- Result: $F(3) = \text{"-"} + 3 + F(1) + 3$

4. Fourth Call: $F(1)$:

- Compute $F(1-3) = F(-2) = \text{"-"}$ (base case)
- Compute $F(1-2) = F(-1) = \text{"-"}$ (base case)
- Result: $F(1) = \text{"-"} + 1 + \text{"-"} + 1 = \text{"-1-1"}$

Substituting Back:

- $F(1) = \text{"-1-1"}$
- $F(3) = \text{"-"} + 3 + \text{"-1-1"} + 3 = \text{"-3-1-13"}$
- $F(2) = \text{"-2-2"}$
- $F(5) = \text{"-2-2"} + 5 + \text{"-3-1-13"} + 5 = \text{"-2-25-3-1-135"}$

Q4 Text Solution:

-> JNDI means Java Naming and Directory Interface is an API that provides naming and directory functionality to apps written using java. It is defined to be independent of any specific directory service implementation.

-> RMI means Remote Method Invocation. It is a technique that enables an object in one system to invoke an object in another JVM.

-> JDK is a bundle of software development tools and supporting libraries combined with the Java Runtime Environment and JVM.

-> JRE is set of components to create and run a java application. A JRE is part of JDK. A JRE is made up of JVM.

Q5 Text Solution:

The correct option is A.

-> Imperative Paradigm is a software development paradigm where functions are coded in every step required to solve a problem. So this is command-based and procedural.

-> Object-oriented programming is imperative statement-based & has abstract data types.

-> Any program written in a logic programming language is a logical set of sentences expressing facts and rules about a problem domain.

-> Functional programming is declarative in nature, involves expression evaluation, and side effects free.

Q6 Text Solution:

Correct option is B.

To calculate the total hard disk space required, let's break it down:

1. Black and white images:

-> Each image is 1-megapixel.

-> Black and white images use 1 bit per pixel.

-> 1 megabyte is 1,048,576 bytes, and 1 byte is 8 bits.

so for each black and white images:

-> 1-megapixel = 1,000,000 pixels.

-> Each pixel requires 1 bit, so each image uses 1,000,000 bits.

-> Converting bits to bytes: $1,000,000/8 = 1,25,000$ bytes = 125 KB.

For 8 black and white images:

-> $125 \text{ KB} * 8 = 1,000 \text{ KB} = 1 \text{ MB}$.

2. 8-color image:

-> The image is 8 megapixels.

-> 8 color means 3 bits per pixel.

-> 8 megapixels = 8,000,000 pixels.



so for the 8 color image:

-> 8 megapixels * 3 bits per pixel = 24,000,000 bits.

-> Converting bits to bytes = 24,000,000/8 = 3,000,000 bytes = 3 MB.

Total hard disk space:

-> Space for 8 black and white images 1 MB.

-> Space for one 8 - 8-color image = 3 MB

total = 1 MB + 3 MB = 4 MB.

Q7 Text Solution:

The correct option is D.

Points A, B, and C are the Bresenham algorithm's characteristics but It is a line-drawing algorithm that determines the points of an n-dimensional raster that should be selected in order to form a close approximation to a straight line between two points not exact.

Q8 Text Solution:

Option A is Correct

Square Vertices (Original Points) $P_0 = (1,1)$, $P_1 = (-1,1)$, $P_2 = (-1,-1)$, $P_3 = (1,-1)$

Parallelogram Vertex (Transferred Points) $Q_0 = (2,1)$, $Q_1 = (0,1)$, $Q_2 = (-2,-1)$, $Q_3 = (0,-1)$

We need to determine the transformation matrix M such that:

$M \cdot P_i = Q_i$ for $i=0,1,2,3$.

The transformation matrix M has the general form:

$$\begin{bmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M = \begin{bmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The coordinates of the transformed points are related to the original points by the transformation:

$$x' = x$$

$$y' = M \cdot y$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Substitute the vertices P_0, P_1, P_2, P_3 and their corresponding transformed vertices Q_0, Q_1, Q_2, Q_3 .

For $P_0=(1,1) \rightarrow Q_0=(2,1)$:

$$\begin{bmatrix} 2 & a & b & 0 & 1 \\ 1 & = & c & d & 0 & . & 1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\text{So, } a+b=2 \dots\dots\dots (1)$$

$$\text{and } c+d=1 \dots\dots\dots (2)$$

$$\text{For } P_1=(-1,1) \rightarrow Q_1=(0,1):$$

$$\begin{bmatrix} 0 & a & b & 0 & -1 \\ 1 & = & c & d & 0 & . & 1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\text{So, } -a+b=0 \dots\dots\dots (3)$$

$$\text{and } -c+d=1 \dots\dots\dots (4)$$

$$\text{For } P_2=(-1,-1) \rightarrow Q_2=(-2,-1):$$

$$\begin{bmatrix} -2 & a & b & 0 & -1 \\ -1 & = & c & d & 0 & . & -1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\text{So, } -a-b=-2 \dots\dots\dots (5)$$

$$\text{and } -c-d=-1 \dots\dots\dots (6)$$

$$\text{For } P_3=(1,-1) \rightarrow Q_3=(0,-1):$$

$$\begin{bmatrix} 0 & a & b & 0 & 1 \\ -1 & = & c & d & 0 & . & -1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\text{So, } a-b=0 \dots\dots\dots (7)$$

$$\text{and } c-d=-1 \dots\dots\dots (8)$$

$$\text{Solving these Equations (1), (2), (3), (4), (5), (6), (7) and (8)}$$

$$a = 1, b = 1, c = 0, d = 1.$$

$$\text{The transformation matrix is:}$$

$$M = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{So, } a-b=0 \dots\dots\dots (7)$$

$$\text{and } c-d=-1 \dots\dots\dots (8)$$

$$\text{Solving these Equations (1), (2), (3), (4), (5), (6), (7) and (8)}$$

$$a = 1, b = 1, c = 0, d = 1.$$

$$\text{The transformation matrix is:}$$

$$M = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Q9 Text Solution:

Option A is Correct

A Bézier curve has certain key properties related to its degree, evaluation, and whether it remains within the convex hull of its control points.

Statement A.

A Bézier curve defined by $n+1$ control points is of degree n . Here, there are 4 control points (P_0, P_1, P_2, P_3), so the curve has degree 3. Hence, Statement A is correct.

Statement B.

The Bézier curve is defined as:

$$P(t) = \sum_{i=0}^3 B_i(t) \cdot P_i,$$



where $B_i(t)$ are the Bernstein polynomials given by:

$$B_i(t) = \binom{3}{i} t^i (1-t)^{3-i}.$$

At $t = \frac{1}{2}$:

$$B_0\left(\frac{1}{2}\right) = \binom{3}{0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^3 = \frac{1}{8},$$

$$B_1\left(\frac{1}{2}\right) = \binom{3}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^2 = \frac{3}{8},$$

$$B_2\left(\frac{1}{2}\right) = \binom{3}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^1 = \frac{3}{8},$$

$$B_3\left(\frac{1}{2}\right) = \binom{3}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^0 = \frac{1}{8}.$$

The curve is then computed as:

$$P\left(\frac{1}{2}\right) = B_0\left(\frac{1}{2}\right) P_0 + B_1\left(\frac{1}{2}\right) P_1 + B_2\left(\frac{1}{2}\right) P_2 + B_3\left(\frac{1}{2}\right) P_3$$

$$P\left(\frac{1}{2}\right) = \frac{1}{8} (-2, 0) + \frac{3}{8} (-2, 4) + \frac{3}{8} (2, 4) + \frac{1}{8} (2, 0)$$

$$(2, 4) + \frac{1}{8} (2, 0)$$

Calculating x-coordinate:

$$x = \frac{1}{8} \times (-2) + \frac{3}{8} \times (-2) + \frac{3}{8} \times 2 + \frac{1}{8} \times 2 = 0$$

$$(2) + \frac{1}{8} \times 2 = 0$$

Calculating y-coordinate:

$$y = \frac{1}{8} \times (0) + \frac{3}{8} \times (4) + \frac{3}{8} \times (4) + \frac{1}{8} \times (0) = 3$$

$$\times (0) = 3$$

Thus, $P\left(\frac{1}{2}\right) = (0, 3)$. Hence Statement B is Correct.

Statement C.

A Bézier curve always lies within the convex hull of its control points. It cannot extend outside the convex hull. Hence, Statement C is incorrect.

Q10 Text Solution:

The correct option is D.

Statement 1 is false, because the maximum number of sides that a triangle might have when clipped to a rectangular viewport is 7.

Statement 2 is true, because in 3D graphics, A linear projection in which 3D objects are projected on a picture plane is called perspective projection. As a result, faraway items look smaller than objects closer to the viewer. Perspective transformation is a non-linear transformation and the perspective transformation is nonlinear in z.

Hence the first statement is false and the second one is true.

Q11 Text Solution:

Correct option is B.

In Beta testing, the software is tested by a larger group of users, typically outside of the organization that developed it.

Q12 Text Solution:

Option A is correct.

MVC (Model View Controller) is an architecture pattern that separates the app structure into three different layers - the Model, the View, and the Controller.

The View : This component is used to work on the UI (user interface) logic of the application.

Q13 Text Solution:

Option C is correct.

In software engineering, Formal methods predominantly use Mathematical notations.

Q14 Text Solution:

option c is correct

If every requirement stated in the SRS has only one interpretation then SRS is said to be unambiguous.

Q15 Text Solution:

Correct option is A.

Availability = Uptime / (Uptime + Downtime)

uptime is the amount of time that the system is operational, and Downtime is the amount of time that the system is not operational.

uptime = 99.99% , MTBF = 1 day

Downtime = MTBF * (1 - Availability).

so the Downtime = 1 day * (1 - 0.9999) = 0.0001 days = 8.64 seconds

To reach the availability goal, the system must repair itself within 9 seconds (approx.) after a failure.

Q16 Text Solution:

Correct option is A.

Code files: These include the source code files that make up the software system. It is necessary to include the entire codebase in the repository



so that the entire system can be rebuilt and deployed if necessary so it need to be committed.

Documentation files: These include documentation that describes how system is designed, how it works, and how to use it. This documentation should be stored in the repository so that it can be easily accessed and understood by developers and other stakeholders so it need to be committed.

Q17 Text Solution:

List I	List II	Explanation
A. Waterfall Model	IV. Inflexible partitioning of the project into stages	The Waterfall Model involves a linear and sequential approach, with inflexible partitioning of the project into stages.
B. Evolutionary Model	I. The software can be developed incrementally	The Evolutionary Model allows for incremental development where the software is developed in iterations.
C. Component-based Software Engineering	II. Requirement compromises are inevitable	In Component-based Software

		Engineering, requirements compromises are inevitable due to the reuse of pre-built components.
D. Spiral Development	III. Explicit recognition of risk	The Spiral Model explicitly includes risk assessment and management at each stage.

Correct Answer: Option D

Q18 Text Solution:

The **Capability Maturity Model (CMM)** defines five levels of process maturity for an organization's software processes, ordered from the least mature to the most mature:

Initial (C) - Processes are ad-hoc and chaotic.

Repeatable (E) - Basic project management processes are in place.

Defined (A) - Processes are well-documented, standardized, and integrated into the organization.

Managed (D) - Processes are measured and controlled.

Optimizing (B) - Focus on continuous process improvement.

Option D is correct.

Q19 Text Solution:

Statement I

- **True:** The Cleanroom model checks the quality of code regularly using statistical methods,



ensuring each part meets high standards before adding it to the system.

Statement II

- **False:** Cleanroom doesn't follow the traditional coding and debugging cycle. Instead, it focuses on **preventing defects** from happening in the first place, rather than finding and fixing them later.

Option C is correct.

Q20 Text Solution:

Assertion (A)

True: In practice, developers rarely perform exhaustive testing (testing every possible input) because it is impractical for all.

Reason (R)

True: Exhaustive testing requires testing all possible input combinations, which grows exponentially with input size, making it computationally expensive and time-consuming.

Relationship Between A and R

The reason (R) explains why exhaustive testing is not feasible in practice, which directly supports the assertion (A).

Option A is correct: Both statements A and R are true, and R is the correct explanation of A.

Q21 Text Solution:

Option A

$\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$

Simplify:

$\neg p \rightarrow (q \rightarrow r)$:

Implication rule, $q \rightarrow r$ is equivalent to $\neg q \vee r$.

So, $\neg p \rightarrow (q \rightarrow r)$ becomes: $\neg p \rightarrow (\neg q \vee r)$

Using the implication rule: $p \vee (\neg q \vee r)$

Simplified: $p \vee \neg q \vee r$

$q \rightarrow (p \vee r)$:

Using the implication rule: $\neg q \vee (p \vee r)$

Simplifies to: $p \vee r \vee \neg q$

Since both parts simplify to the same expression ($p \vee r \vee \neg q$), **Option A is logically equivalent.**

Q22 Text Solution:

Statement I

$\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + (1/2) \rfloor$ for all real number x

True:

When we multiply x by 2, the floor of the split is based on the integer and fractional parts of x.

suppose $x = 1, 2, 3 - - - -$

Take $x = 1$

$$\lfloor 2 \times 1 \rfloor = \lfloor 1 \rfloor + \lfloor 1 + (1/2) \rfloor$$

$$\lfloor 2 \rfloor = \lfloor 1 \rfloor + \lfloor (3/2) \rfloor$$

$$\lfloor 2 \rfloor = \lfloor 1 \rfloor + \lfloor 1.5 \rfloor$$

$$2 = 1 + 1$$

$$2 = 2$$

Statement II

$\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$ for all real numbers x and y

False:

The floor of x + y is not always equal to the sum of the individual floors of x and y. For instance, if $x = 0.5$ and $y = 0.5$, then $\lfloor x + y \rfloor = \lfloor 1 \rfloor = 1$, but $\lfloor x \rfloor + \lfloor y \rfloor = 0 + 0 = 0$.

Option C is correct.

Q23 Text Solution:

Determine the Distribution of Jobs

Since there are 5 jobs and 4 employees, and every employee must get at least one job, there are two possible distributions of jobs among employees:

One employee gets 2 jobs, and each of the other three employees gets 1 job.

Calculate the Number of Ways for Each Case

Case 1: One employee gets 2 jobs, and each of the other three employees gets 1 job.

Choose the employee who will get 2 jobs: There are 4 ways to choose this employee.

Choose 2 jobs out of 5 to give to this

employee: There are $\binom{5}{2} = 10$ ways to choose 2 jobs out of 5.

Assign the remaining 3 jobs to the other 3

employees: The remaining 3 jobs can be assigned to 3 employees in $3! = 6$ ways.

Using the multiplication rule, the total number of ways for this case is:

$$4 \times 10 \times 6 = 240$$

Option C is correct.

Q24 Text Solution:

Calculate the Total Number of Bins



The warehouse has:

50 aisles

85 horizontal locations per aisle

5 shelves

So, the total number of bins is given by:

$$\text{Total bins} = 50 \times 85 \times 5 = 21250$$

This means there are **21250 unique bins** available.

According to the **Pigeonhole Principle**, if we want **at least two products to be stored in the same bin**, so, one more product than the total number of bins.

The least number of products required so that at least two products must be stored in the same bin is:

$$21250 + 1 = 21251$$

Option D is correct.

Q25 Text Solution:

Let $f(n)$ be the number of ways to reach the n^{th} stair. The recurrence relation is:

$$f(n) = f(n-1) + f(n-2)$$

- If the person takes 1 step, they have $f(n-1)$ ways to reach the remaining stairs.
- If the person takes 2 steps, they have $f(n-2)$ ways to reach the remaining stairs.

The base cases are:

- $f(1) = 1$: There is 1 way to reach the first stair.
- $f(2) = 2$: There are 2 ways to reach the second stair.

Calculate $f(n)$ for $n=3$ to $n=8$:

1. $f(3) = f(2) + f(1) = 2 + 1 = 3$
2. $f(4) = f(3) + f(2) = 3 + 2 = 5$
3. $f(5) = f(4) + f(3) = 5 + 3 = 8$
4. $f(6) = f(5) + f(4) = 8 + 5 = 13$
5. $f(7) = f(6) + f(5) = 13 + 8 = 21$
6. $f(8) = f(7) + f(6) = 21 + 13 = 34$

So, the number of ways to climb a flight of 8 stairs is **34**.

Option D is correct.

Q26 Text Solution:

A Wheel graph W_n is created by adding a central vertex to a cycle graph C_{n-1} and connecting this central vertex to all other vertices in the cycle.

A **regular graph** is a graph in which each vertex has the same degree.

The central vertex in W_n has a degree of $n-1$ (it is connected to all vertices in the cycle).

Each vertex in the outer cycle has a degree of 3: two edges connecting to adjacent vertices in the cycle and one edge connecting to the center.

Condition for Regularity

For W_n to be regular, the degrees of all vertices (both the central and the cycle vertices) must be the same.

This is when $n-1 = 3$, means that the central vertex has the same degree as the outer vertices.

Solving for n

$$n-1 = 3 \text{ i.e. } n=4$$

Option C is correct.

Q27 Text Solution:

Graph A:

Graph A has six vertices it can be drawn without edges crossing.

It resembles K_5 (the complete graph with five vertices) without any crossings. Therefore, **Graph A is planar**.

Graph B:

Graph B includes a complete bipartite subgraph $K_{3,3}$ structure, with six vertices divided into two sets of three, each connected to all vertices in the other set.

The graph $K_{3,3}$ is known to be **non-planar** (by Kuratowski's theorem).

Therefore, **Graph B is not planar**.

Graph C:

Graph C to have a structure that includes a complete bipartite subgraph or additional complex connections.

This structure creates edge crossings that cannot be eliminated by redrawing, indicating a **non-planar** structure.

Thus, **Graph C is not planar**.

Option C is correct.



Q28 Text Solution:

List I	List II	Explanation
Identity	Name	
A. $x+x=x$	III. Idempotent Law	This is the Idempotent Law where $x+x$ simplifies to x .
B. $x+0=x$	I. Identity Law	This is the Identity Law , stating that adding zero to x yields x itself.
C. $x+1=1$	IV. Domination Law	This is the Domination Law where any term added to 1 (true) in boolean algebra results in 1.
D. $x+xy=x$	II. Absorption Law	This is the Absorption Law , as x absorbs xy and simplifies to x .

Correct Answer: Option C**Q29 Text Solution:**Statement A: $a*a=a$ for every a in X .

True: This statement implies that every element in X is idempotent, means that applying the operation on the same element twice yields the element itself.

In many semigroups, this property can hold, if the operation is idempotent by design.

Statement B: $a*b*a=a$ for every a, b in X .

True: This statement implies a form of absorption, where surrounding b with a results in a .

In algebraic structures with specific properties, this type of identity is present.

Statement C: $a*b*c=a*c$ for every a, b, c in X .

True: This statement implies that b does not affect the outcome of the operation with a and c , true in specific semigroup structures where intermediate elements do not change the result. Option D is correct.

Q30 Text Solution:**Step 1: Convert Inequalities to Equations**

To identify the boundary lines of the feasible region:

$$2x-3y=5$$

$$x+3y=11$$

$$4x+y=15$$

$$x=0$$

$$y=0$$

Step 2: Find Intersection Points (Vertices)

To determine the feasible region, Each intersection point is a potential vertex of the feasible region.

Intersection of $2x-3y=5$ and $x+3y=11$:

From $x+3y=11$, solve for x :

$$x=11-3y$$

Substitute $x=11-3y$ into $2x-3y=5$:

$$2(11-3y)-3y=5$$

$$22-6y-3y=5$$

$$22-9y=5$$

$$9y=17 \Rightarrow y=17/9 \approx 1.89$$

Substitute $y=1.89$ back into $x=11-3y$:

$$x=11-3 \times 1.89 \approx 5.33$$

So, one vertex is approximately $(5.33, 1.89)$.

Intersection of $2x-3y=5$ and $4x+y=15$:

From $4x+y=15$, solve for y :

$$y=15-4x$$

Substitute $y=15-4x$ into $2x-3y=5$:

$$2x-3(15-4x)=5$$

$$2x-45+12x=5$$

$$14x=50$$

$$x=50/14 \approx 3.57$$

Substitute $x=3.57$ back into $y=15-4x$:

$$y=15-4 \times 3.57 \approx 0.72$$

So, another vertex is approximately $(3.57, 0.72)$.

Intersection of $x+3y=11$ and $4x+y=15$

From $x+3y=11$, solve for x :

$$x=11-3y$$



Substitute $x=11-3y$ into $4x+y=15$:

$$4(11-3y)+y=15$$

$$44-12y+y=15$$

$$44-11y=15$$

$$11y=29$$

$$y=29/11 \approx 2.64$$

Substitute $y = 2.64$ back into $x = 11-3y$:

$x = 11- 3 \times 2.64 \approx 3.08$ So, another vertex is approximately (3.08,2.64).

Intersection with the Axes

For the intersections with the x- and y-axes:

When $x=0$:

$$\text{From } x+3y = 11:$$

$$y=11/3 \approx 3.67$$

$$\text{From } 4x + y = 15: y=15$$

So, (0,3.67) and (0,15) are additional points, but only (0,3.67) is feasible within the constraints.

When $y=0$:

$$\text{From } 2x - 3y = 5:$$

$$x = 5/2 = 2.5$$

$$\text{From } x + 3y = 11:$$

$$x=11$$

So, (2.5,0) and (11,0) are additional points, but only (2.5,0) is feasible within the constraints.

Step 3: Evaluate the Objective Function at Each Vertex

Evaluate $Z = 6x+5y$ at each feasible vertex:

At (5.33,1.89):

$$Z = 6 \times 5.33 + 5 \times 1.89 \approx 31.72$$

At (3.57,0.72):

$$Z = 6 \times 3.57 + 5 \times 0.72 \approx 25$$

At (3.08,2.64):

$$Z = 6 \times 3.08 + 5 \times 2.64 \approx 31.72$$

At (2.5,0):

$$Z = 6 \times 2.5 + 5 \times 0 = 15$$

At (0,3.67):

$$Z = 6 \times 0 + 5 \times 3.67 \approx 18.35$$

Step 4: Determine the Optimal Solution

The maximum value of Z is **31.72**, which occurs at both points (5.33,1.89) and (3.08,2.64).

Option C is correct.

Q31 Text Solution:

Language A: $L = \{(01)^n 0^k \mid n > k, k \geq 0\}$

- This language contains strings with a repeating pattern $(01)^n$ followed by 0^k , where $n > k$.
- This language is not regular because a regular language cannot count and compare the numbers n and k .
- The requirement $n > k$ shows a dependency between n and k , which a regular language cannot handle. ie means not regular.

Language B: $L = \{c^n b^k a^{n+k} \mid n \geq 0, k \geq 0\}$

- This language has strings of the $c^n b^k a^{n+k}$ where the number of a symbols is the sum of the number of c and b symbols.
- This language is not regular because regular languages cannot handle the dependency a^{n+k} .
- To recognize this language, a machine would need to "remember" the count of both c's and b's to verify the count of a's, which requires more memory than a finite automaton provides. ie means not regular.

Language C: $L = \{0^n 1^k \mid n \neq k\}$

- This language consists of strings of 0's followed by 1's, where the number of 0's is not equal to the number of 1's.
- This language is also not regular. Regular languages can typically handle equality conditions (like $n=k$ in the language $\{0^n 1^n\}$), but not inequality conditions like $n \neq k$, which would require counting the exact number of symbols. ie means not regular.

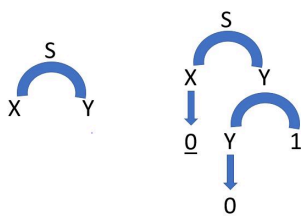
The Final Answer is: A, B, and C

Q32 Text Solution:

Explanation:

The Context-free grammar has at least two 0's as shown in below figure



**Q33 Text Solution:**

Explanation:

$$L_3 = L_1 \cap L_2$$

$$L_1 = \{0^n 1^n 0^m \mid n \geq 1, m \geq 1\}$$

$$= \{010, 0100, 00110, 001100, \dots\}$$

$$L_2 = \{0^n 1^m 0^m \mid n \geq 1, m \geq 1\}$$

$$= \{010, 0100, 01100, 001100, \dots\}$$

$$L_3 = \{0^n 1^n 0^n \mid n \geq 1\}$$

$$= \{010, 001100, 000111000, \dots\}$$

Here L_3 is intersection of L_1 and L_2

So statement A is correct.

Statement B: L_1 is context-free: A context-free grammar can generate strings where the first n zeros are followed by n ones, and then any number of zeros. The structure of the first two segments can be handled by a simple stack-based machine (like a pushdown automaton), and the final segment (the extra zeros) does not require any specific relation to the previous segments. So, it is context-free.

L_2 is a context-free language. This language can be generated by context-free grammar. The number of zeros in the third segment must match the number of ones in the second segment, which is a classic context-free pattern (e.g., matching the number of ones and zeros). Again, this can be handled by a pushdown automaton, making L_2 context-free.

L_3 is not context-free. This language requires that the number of zeros in the first segment matches the number of ones in the second segment and the number of zeros in the third segment. This is a non-context-free condition because context-free grammar cannot enforce the condition that the first, second, and third segments all have the same length.

Statement D: L_1 is a subset of L_3

$$L_1 = \{0^n 1^n 0^m \mid n \geq 1, m \geq 1\}$$

$$= \{010, 0100, 00110, 001100, \dots\}$$

$$L_3 = \{0^n 1^n 0^n \mid n \geq 1\}$$

$$= \{010, 001100, 000111000, \dots\}$$

As see, L_1 is superset L_3

Statement D is incorrect.

Q34 Text Solution:

Explanation:

True: Applying a homomorphism (a function that maps each symbol in the alphabet of a context-free language to a string over another alphabet), the result is still a context-free language.

True: If we take the reversal of any context-free language (reversing each string in the language), the resulting language is still context-free.

The Final Answer is: Both Statement I and Statement II are true

Q35 Text Solution:

Option C is Correct

Step: 1 The diagram contains the following states and transition table:

a, b, c, d, e, f, g, h

	0	1
a	b	f
b	g	*c
*c	a	*c
e	h	f
f	*c	g
g	g	e
h	g	*c

Step 2: Delete all states that aren't reached from the initial state to the final state.

state (d) is that one state.

step 3:- Find equivalence set

- equivalent set:- separate final and Non. final

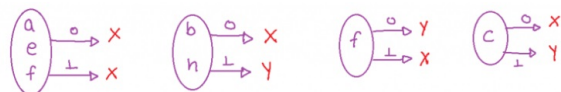


state

$x \{a \ b \ e \ f \ g \ h\} y \{c\}$

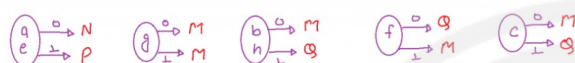
1: equivalent set:- take 0 equivalent sets and see each state transition on input 0 and 1. If reaches the same set then it is also 1 equivalent set otherwise make a different set.

$M[a \ e \ g] \ N[b \ h] \ P[f] \ Q[c]$

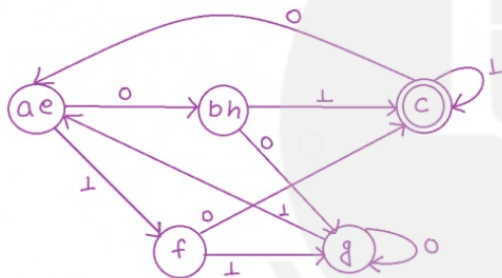


2 equivalent sets: Take 1 equivalent set and see each state transition on input 0 and 1. If you reach the same set, then it is also 1 equivalent set; otherwise, make a different set.

$[a \ e] \ [b \ h] \ [f] \ [c]$



The automaton cannot be minimized further. Therefore, the minimum number of states required is: 5 States



Q36 Text Solution:

Option B is Correct

We are given a finite automaton with three states, for better understanding Suppose:

- **Start state (A):** The leftmost state.
- **Intermediate state (B):** The middle state.
- **Final state (C):** The rightmost state.

Initial Transitions (A to B): Starting at A, the automaton can transition to B on a or b. This is expressed as: $(a+b)$

Transition from B to C: At B, to transition to C, the automaton requires ab (a sequence of a followed by b). This is expressed as: ab

Behavior at State C: At C, the automaton allows Loops on a or b, which can be expressed as

$(a+b)^*$. Transitions back to B via a, followed by further sequences from B (e.g., ab).

The automaton accepts strings that:

1. Start with $(a+b)$,
2. Follow with ab,
3. Continue with repeated patterns involving:

- ab (returning to C),
- bb (looping on C),
- $aa^*(a+b)ab(a+b)ab$ (complex sequences involving C and returning to B).

This is expressed as:

$(a+b)ab(ab+bb+aa^*(a+b)ab)^*$

Q37 Text Solution:

• Initial State and Acceptance:

- The automaton starts in state q_0 and q_1 aims to reach the accepting state q_t .

• Input and Stack Alphabets:

- **Input alphabet:** $\{a,b\}$.
- **Stack alphabet:** $\{a,b,z\}$ (where z is the initial stack symbol).

• Language Description:

- The language L accepted by M consists of strings in the form ww^R , where w is any non-empty string over $\{a,b\}$, and w^R , represents the reverse of w.

• Behavior of the PDA:

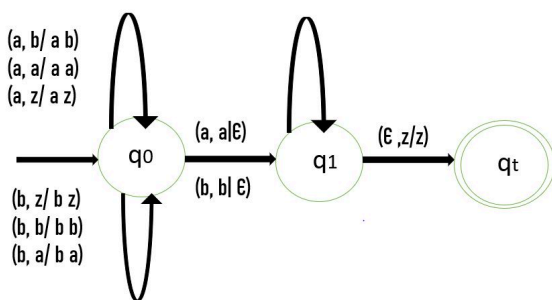
- The automaton reads the first half of the string, pushes symbols onto the stack, then matches the second half in reverse by popping the stack.

• Accepted Language:

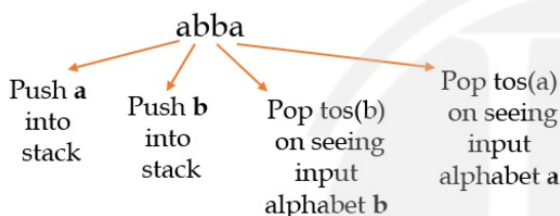
- The language accepted by M is $L = \{ww^R \mid w \in \{a,b\}^+\}$, representing palindromic strings where the second half mirrors the first half.

The push down automata for the given transition is :





First, keep pushing any number of a's and b's in the stack, and then pop one a from the stack on seeing input alphabet a and pop one b from stack on seeing input b. Keep popping until we reach end of string ϵ . when we reach end of string ϵ and Top of Stack is z, then go to final state. Let's take one string and analyze them



Q38 Text Solution:

List I	List II
Production Rules	Grammar
A. $S \rightarrow XY$ $X \rightarrow 0$ $Y \rightarrow 1$	III. Chomsky Normal Form
B. $S \rightarrow aS \mid bSS \mid c$	IV. S. Grammar
C. $S \rightarrow AB$ $A \rightarrow 0A \mid 1A \mid 0$ $B \rightarrow 0A$	I. Greibach Normal Form
D. $S \rightarrow aAbc$ $Ab \rightarrow bA$ $Ac \rightarrow Bbcc$ $bB \rightarrow Bb$ $aB \rightarrow aa \mid aaA$	II. Context sensitive Grammar

The Final Answer is: A - III, B - IV, C - I, D - II

Q39 Text Solution:

Explanation:

Depth First Search (DFS) is a method used to traverse or search a graph by exploring as deeply as possible along each path before backtracking.

DFS is valuable for tasks like cycle detection, topological sorting, and analyzing graph structures.

Dominators: A node u is said to dominate a node v (for a source node S) if every path from S to v passes through u . In simpler terms, u is unavoidable on any route from S to v . This concept is commonly used in program optimization and analysis of control flow graphs. A reducible graph is a control flow graph (CFG) where loops are well-structured and can be simplified into standard forms.

The Final Answers are A, B, and C.

Q40 Text Solution:

LL(1) is a Top-down parsing technique, where parsing starts from the root and works down towards the leaves of the parse tree.

LR is a Bottom-up parsing technique. It starts from the leaves (input tokens) and works towards the root of the parse tree

Recursive descent parser is indeed a Top-down parser. It works by recursively applying the production rules starting from the start symbol. SLR (Simple LR) is a Bottom-up parser, which is an improved version of the basic LR parsing technique. It uses a stack to handle parsing in a bottom-up manner.

The Final Answer is that both Statement I and Statement II are false

Q41 Text Solution:

$$\begin{aligned}
 &\text{Given } (A+B) * (C * D - E) * F / G \\
 &= (AB+) * (\underline{C * D} - E) * F / G \\
 &= (AB+) * (CD * -E) * F / G \\
 &= (AB+) * (\underline{CD * E-}) * F / G \\
 &= (\underline{AB+}) * (CD * E-) * F / G \\
 &= (AB+ CD * E- *) * F / G \\
 &= (AB+ \underline{CD * E- * F*}) / G \\
 &= (AB+ \underline{CD * E- * F*} G /)
 \end{aligned}$$

Q42 Text Solution:

Explanation:



A Dequeue (Double-ended Queue) allows adding and removing items from both ends (front and rear).

A Stack operates on a Last In, First Out (LIFO) principle, with the following operations: 1. Push
2. Pop

The time complexity for AddFront in a Dequeue is $O(1)$, as it is similar to the PUSH operation in a stack. However, for AddRear, using a stack requires transferring all elements to another stack (using POP and PUSH), making the time complexity $O(n)$.

The Final Answer is: $O(1)$ and $O(n)$.

Q43 Text Solution:

Explanation:

To merge two balanced binary trees with m and n elements into a balanced binary search tree.

Perform an In-order traversal on each tree to get two sorted arrays, taking $O(m)$ and $O(n)$ time, respectively.

Merge the two sorted arrays into a single sorted array in $O(m+n)$ time.

So, we can build a balanced BST from the merged array in $O(m+n)$ time.

The final answer is $O(m+n)$.

Q44 Text Solution:

Explanation:

In a balanced BST (e.g., AVL tree or Red-Black tree), elements are stored in a sorted order.

The smallest element is always the leftmost node in a BST, so finding it takes $O(\log n)$ in a balanced tree. Deleting it also takes $O(\log n)$ because re-balancing the tree might be required. A BST satisfies both condition.

In a min-heap, the smallest element is at the root, so deletion of the smallest element takes $O(\log n)$.

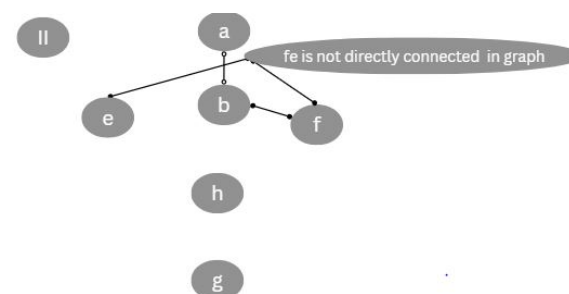
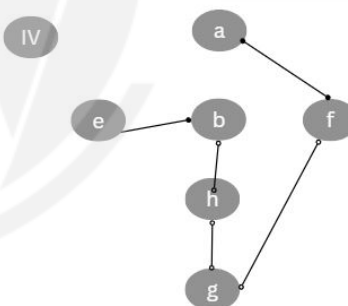
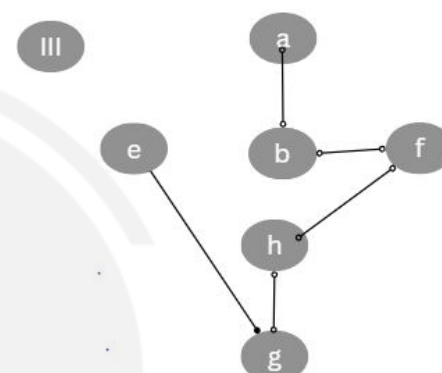
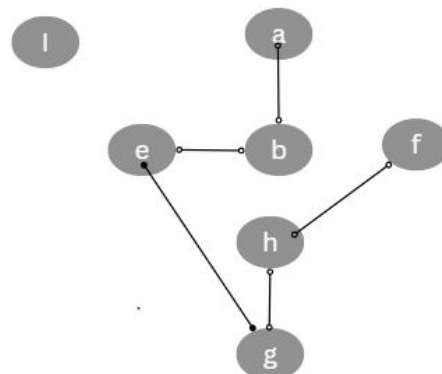
ie, a heap does not satisfy the insertion condition. To insert an element in Heap, if it is not present in the set, we need to search the element first. To search the element the time complexity is $O(n)$.

The Final Answer is: A balanced binary search tree can be used but not a heap.

Q45 Text Solution:

Explanation:

DFS is used in Stack.



Sequence: (a b e g h f, a b f h g e, a f g h b e)

The Final Answer is: I, III, and IV only.

Q46 Text Solution:

The correct answer is option A.



(True) Statement I: In an undirected graph, the number of vertices with an odd degree is even. This is a well-known result in graph theory, often referred to as the Handshaking Lemma. The lemma states that the sum of the degrees of all vertices in a graph is equal to twice the number of edges. Since each edge contributes 2 to the total degree count, the sum of the degrees is even. The vertices with odd degrees must pair up to contribute to this even sum, meaning that the number of vertices with an odd degree must be even.

(True) Statement II: In an undirected graph, the sum of the degrees of all vertices is even. This is also true, due to the Handshaking Lemma. The sum of the degrees is equal to twice the number of edges, which is always even.

Q47 Text Solution:

The correct answer is **option A**.

(A) ($f_1(n) = 2^n$): This is an exponential function, which grows very quickly as (n) increases. Its complexity is ($O(2^n)$).

(B) ($f_2(n) = n^{3/2}$): This is a polynomial function with a degree of (1.5). Its complexity is ($O(n^{3/2})$).

(C) ($f_3(n) = n \log n$): This is a function that grows faster than linear but slower than any polynomial of degree greater than 1. Its complexity is ($O(n \log n)$).

(D) ($f_4(n) = n^{\log n}$): This function grows faster than any polynomial function of fixed degree but slower than exponential functions. Its complexity is ($O(n^{\log n})$).

Note that ($n^{\log n}$) grows faster than ($n^{3/2}$) for sufficiently large (n).

Thus, the increasing order of growth rates is:

($C < B < D < A$)

Q48 Text Solution:

The order of a leaf node in a B+ tree $\Rightarrow m*(k+r) + b \leq \text{Block Size}$

where m = order of B+ tree, k = value, r = record pointer, b = block pointer

Given:

Block size = 1K bytes = 1024 bytes

Size of a data record pointer (r) = 7 bytes

Size of a value field (k) = 9 bytes

Size of block pointer (b) = 6 Bytes

$m*(k+r) + b = \text{Block Size}$

$m*(9+7) + 6 = 1024$

$16m = 1018$

$m = 63$ (approx)

Q49 Text Solution:

The correct answer is **option C**.

To determine the location of key 18 using the hash function ($h(\text{key}) = \text{key} \bmod 7$) with linear probing, Here are the following steps:

1. Calculate the hash for each key.
2. Insert the keys into the table using linear probing if there are collisions.

Now calculate the hash values for each key:

Key 44: ($h(44) = 44 \bmod 7 = 2$) Place 44 at index 2.

Key 45: ($h(45) = 45 \bmod 7 = 3$) Place 45 at index 3.

Key 79: ($h(79) = 79 \bmod 7 = 2$) Index 2 is occupied (by 44), so we probe to the next index (3 is also occupied by 45). Index 4 is free, so place 79 at index 4.

Key 55: ($h(55) = 55 \bmod 7 = 6$) Place 55 at index 6.

Key 91: ($h(91) = 91 \bmod 7 = 0$) Place 91 at index 0.

Key 18: ($h(18) = 18 \bmod 7 = 4$) Index 4 is occupied (by 79), so we probe to the next index (5 is free).

Place 18 at index 5.

Key 63: ($h(63) = 63 \bmod 7 = 0$) Index 0 is occupied (by 91), so we probe to the next indices (1 is free). Place 63 at index 1.

Now, the final positions in the table:

Index 0: 91

Index 1: 63

Index 2: 44

Index 3: 45

Index 4: 79

Index 5: 18



Index 6: 55

Thus, the location of key 18 is 5.

Q50 Text Solution:

The correct answer is **option A**.

A. Bellman-Ford algorithm: The time complexity is $O(n \cdot m)$. This is because the algorithm relaxes all edges up to $(n-1)$ times, where n is the number of vertices, and m is the number of edges.

B. Kruskal's algorithm: The time complexity is $O(m \cdot \log n)$. This is due to the need to sort the edges, which takes $O(m \cdot \log m)$ time.

C. Floyd-Warshall algorithm: The time complexity is $O(n^3)$. This algorithm uses three nested loops to compute the shortest paths between all pairs of vertices.

D. Topological sorting: The time complexity is $O(n + m)$. This is because it processes each vertex and edge once during the sorting.

Q51 Text Solution:

The correct answer is **option A**.

Utility-based agent: This type of agent evaluates its actions based on a utility function, which assigns a numerical value (utility) to each state. The agent aims to maximize its overall utility, which can be interpreted as achieving "happiness" or "satisfaction" in a given context. In this sense, the agent can differentiate between "happy" (more preferred) and "unhappy" (less preferred) states based on the utility values assigned to them.

Q52 Text Solution:

(True) Statement I: "Breadth-First Search is optimal when all the step costs are equal whereas uniform-cost search is optimal with any step-cost."

- Breadth-First Search (BFS): This search algorithm is optimal when all step costs are equal because it explores all nodes at the present depth before moving on to nodes at the next depth level. This means that it will find the shortest path in terms of the number of edges when all edges have the same cost.

- Uniform-Cost Search: This search algorithm is optimal for any step cost, including cases where step costs vary. It expands the least-cost node first, ensuring that it always finds the least-cost path to the goal.

Statement II is TRUE because at depth d , when we implement breadth first search with same cost at all nodes, it will have to search lesser nodes but uniform cost search will always check for the lowest cumulative cost, hence, will expend more nodes than BFS.

Q53 Text Solution:

The correct answer is **option B**.

Let $\alpha = -\infty$ and $\beta = +\infty$. The initial call starts from node A, with values of α and β passed down the tree.

- At A, the maximizer chooses between B and C, calling B first. At B, the minimizer compares D and E and calls D.
- In D, the left child returns a value of 3, giving $\alpha = \max(-\infty, 3) = 3$. Since β is still $+\infty$, D checks its right child, which returns 5, updating $\alpha = \max(3, 5) = 5$. D returns 5 to B.
- At B, $\beta = \min(+\infty, 5) = 5$. B calls E to see if it can find a lower value.
- At E, with $\alpha = -\infty$ and $\beta = 5$, the left child returns 6, leading to $\alpha = \max(-\infty, 6) = 6$. Now, $\beta \leq \alpha$ is true, so E returns 6 to B without checking the right child.
- B updates $\beta = \min(5, 6) = 5$ and returns 5 to A. A then compares the values from B and C, ensuring it gets at least 5 from B.
- At C, with $\alpha = 5$ and $\beta = +\infty$, it calls F. F checks its left child (1) and right child (2), but the best value remains 5. F returns 2 to C.
- At C, $\beta = \min(+\infty, 2)$ leads to $\beta \leq \alpha$ being true. C exits without further evaluation.
- Finally, A receives 2 from C, leading to the best value being $\max(5, 2) = 5$. Therefore, the optimal value the maximizer can achieve is 5.

Q54 Text Solution:

The correct answer is **option C**.



Statement A: "Greedy best-first search is not optimal but is often efficient."

This statement is true. The greedy best-first search uses a heuristic to guide its search, which can make it efficient in terms of time, but it does not guarantee an optimal solution.

Statement B: "A is complete and optimal provided $h(n)$ is admissible or consistent."*

This statement is true. A^* search is complete (it will find a solution if one exists) and optimal (it will find the least-cost path) if the heuristic ($h(n)$) is admissible (never overestimates the true cost to reach the goal) or consistent (satisfies the triangle inequality).

Statement C: "Recursive best-first search is efficient in terms of time complexity but poor in terms of space complexity."

This statement is false. Recursive best-first search is actually efficient in terms of space complexity because it uses linear space (proportional to the depth of the search tree). However, it can be inefficient in terms of time complexity because it may revisit nodes multiple times.

Statement D: " $h(n) = 0$ is an admissible heuristic for the 8-puzzle."

This statement is true. A heuristic that always returns 0 is admissible because it never overestimates the cost to reach the goal (it estimates it as 0, which is less than or equal to the actual cost).

Q55 Text Solution:

The correct answer is **option A**.

To express the sentence "There is a country that borders both India and Pakistan" using logical expressions, we need to break down the sentence:

Existence of a Country: The expression should indicate that there exists at least one country.
Borders with India and Pakistan: The expression should indicate that this country shares borders

with both India and Pakistan.

Option A: $\exists c \text{ Country}(c) \wedge \text{Border}(c, \text{India}) \wedge \text{Border}(c, \text{Pakistan})$

This expression states that there exists a country (c) such that (c) borders India and (c) borders Pakistan. This correctly expresses the sentence.

Q56. Text Solution:

The correct answer is **option A**.

$$P(\text{cavity}) = P(\text{cavity} \wedge \text{toothache} \wedge \text{catch}) + P(\text{cavity} \wedge \text{toothache} \wedge \neg \text{catch}) + P(\text{cavity} \wedge \neg \text{toothache} \wedge \text{catch}) + P(\text{cavity} \wedge \neg \text{toothache} \wedge \neg \text{catch})$$

Substituting the values from the table:

$$P(\text{cavity}) = 0.108 + 0.012 + 0.072 + 0.008 = 0.200$$

Thus, the marginal probability $P(\text{cavity})$ is 0.200

Q57. Text Solution:

The correct answer is **option B**.

The probability of $P(\text{cavity} | \text{toothache})$ is calculated using Bayes' theorem:

$$P(\text{cavity} | \text{toothache}) = \frac{P(\text{cavity} \wedge \text{toothache})}{P(\text{toothache})}$$

Calculate $P(\text{cavity} \wedge \text{toothache})$

This involves summing the probabilities where both "cavity" and "toothache" are true, regardless of "catch":

$$P(\text{cavity} \wedge \text{toothache}) = P(\text{cavity} \wedge \text{toothache} \wedge \text{catch}) + P(\text{cavity} \wedge \text{toothache} \wedge \neg \text{catch})$$

From the table:

$$P(\text{cavity} \wedge \text{toothache}) = 0.108 + 0.012 = 0.120$$

Calculate $P(\text{toothache})$

This is the marginal probability of "toothache", summing over all probabilities where "toothache" is true (both cavity and $\neg$ cavity):

$$P(\text{toothache}) = P(\text{toothache} \wedge \text{catch}) + P(\text{toothache} \wedge \neg \text{catch})$$

From the table:

$$P(\text{toothache}) = (0.108 + 0.012) + (0.016 + 0.064) = 0.200$$

Calculate $P(\text{cavity} | \text{toothache})$

Substitute into the formula,

$$P(\text{cavity} | \text{toothache}) = \frac{0.120}{0.200} = 0.600$$

The probability of a cavity given a toothache is 0.600.



Q58. Text Solution:

The correct answer is **option B**.

The probability of $P(\text{cavity} | \text{toothache})$ is calculated using **Bayes' theorem**:

$$P(\text{toothache} | \text{cavity}) = \frac{P(\text{toothache} \wedge \text{cavity})}{P(\text{cavity})}$$

Calculate $P(\text{toothache} \wedge \text{cavity})$

This involves summing the probabilities where both "toothache" and "cavity" are true, regardless of "catch":

$$P(\text{toothache} \wedge \text{cavity}) = P(\text{toothache} \wedge \text{cavity} \wedge \text{catch}) + P(\text{toothache} \wedge \text{cavity} \wedge \neg \text{catch})$$

From the table:

$$P(\text{toothache} \wedge \text{cavity}) = 0.108 + 0.012 = 0.120$$

$$P(\text{cavity}) = P(\text{cavity} \wedge \text{toothache} \wedge \text{catch}) +$$

$$P(\text{cavity} \wedge \text{toothache} \wedge \neg \text{catch}) + P(\text{cavity} \wedge \neg \text{toothache} \wedge \text{catch}) + P(\text{cavity} \wedge \neg \text{toothache} \wedge \neg \text{catch})$$

Substituting the values from the table:

$$P(\text{cavity}) = 0.108 + 0.012 + 0.072 + 0.008 = 0.200$$

Thus, the marginal probability $P(\text{cavity})$ is 0.200

Substitute into the formula mentioned,

$$P(\text{toothache} | \text{cavity}) = \frac{0.120}{0.200} = 0.600$$

The probability of a cavity given a toothache is **0.600**

Q59. Text Solution:

The correct answer is **option C**.

To find $P(\text{cavity} \vee \text{toothache})$, we calculate the probability that either **cavity** or **toothache** (or both) occur. This can be found using the

inclusion-exclusion principle:

$$P(\text{cavity} \vee \text{toothache}) = P(\text{cavity}) + P(\text{toothache}) - P(\text{cavity} \wedge \text{toothache})$$

Calculate $P(\text{cavity})$

From a previous calculation: Calculated in Q1

$$P(\text{cavity}) = 0.200$$

Calculate $P(\text{toothache})$

From a previous calculation: Calculated in Q2

$$P(\text{toothache}) = 0.200$$

Calculate $P(\text{cavity} \wedge \text{toothache})$

From a previous calculation: Calculated in Q2

$$P(\text{cavity} \wedge \text{toothache}) = 0.120$$

Apply the Formula

Substitute the values into the inclusion-exclusion formula:

$$P(\text{cavity} \vee \text{toothache}) = P(\text{cavity}) + P(\text{toothache}) - P(\text{cavity} \wedge \text{toothache})$$

$$P(\text{cavity} \vee \text{toothache}) = 0.200 + 0.200 - 0.120 = 0.280$$

The probability $P(\text{cavity} \vee \text{toothache})$ is **0.280**

Q60. Text Solution:

The correct answer is **option D**.

To calculate $P(\text{cavity} | \text{toothache} \vee \text{catch})$, we use the formula for conditional probability:

Compute $P(\text{Cavity} \wedge (\text{Toothache} \vee \text{Catch}))$

$$= P(\text{cavity} | \text{toothache} \vee \text{catch}) = \frac{(P(\text{cavity}) \wedge (\text{toothache} \vee \text{catch}))}{P(\text{toothache} \vee \text{catch})}$$

This is the probability that "cavity" is true and either "toothache" or "catch" is true. In logical terms:

From the table:

1. $P(\text{cavity} \wedge \text{toothache} \wedge \text{catch}) = 0.108$
2. $P(\text{cavity} \wedge \text{toothache} \wedge \neg \text{catch}) = 0.012$
3. $P(\text{Cavity} \wedge \neg \text{Toothache} \wedge \text{Catch}) = 0.072$

Adding these, we get:

$$P(\text{cavity} \wedge (\text{toothache} \vee \text{catch})) = 0.108 + 0.012 + 0.072 = 0.192$$

Calculate $P(\text{Toothache} \vee \text{Catch})$

For $P(\text{Toothache} \vee \text{Catch})$, sum all probabilities where either "toothache" or "catch" is true (both for Cavity and $\neg$ Cavity):

$$P(\text{toothache} \vee \text{catch}) = 0.108 + 0.012 + 0.072 + 0.016 + 0.064 + 0.144 = 0.416$$

Calculate $P(\text{cavity} | \text{toothache} \vee \text{catch}) =$

$$\frac{0.192}{0.416} = 0.4615$$

Now substitute into the formula:

Q61 Text Solution:

The component of a Database Management System (DBMS) that ensures that concurrent execution of multiple operations on the database results in a consistent database state is the Transaction Processing System.

Q62 Text Solution:

The correct answer is **option A**.

Statement I: "Relational database schema represents the logical design of the database."

This statement is TRUE. A relational database schema defines the structure of the database in terms of tables (relations), fields (attributes), and the relationships between them. It represents the logical design of the database, outlining how data is organized and how the different pieces of data relate to each other.

Statement II: "Current snapshot of a relation only provides the degree of the relation."

This statement is FALSE. The current snapshot of a relation (or table) provides both the degree (the number of attributes/columns) and the cardinality (the number of tuples/rows) of the relation. Therefore, it does not only provide the degree.

Q63 Text Solution:

The correct answer is **option D**.

To find the average access time needed to access any record in a fixed-length record file that is ordered on the key field using binary search, we can follow these steps:

Accessing Disk Blocks: In a binary search, the maximum number of comparisons needed to find a record in a sorted list of size (R) is ($\log_2(R)$). However, since we are dealing with disk blocks, we need to consider how many disk accesses are required.

Disk Blocks: Given that the file is stored in (B) disk blocks, and assuming that each block can hold multiple records, we need to determine how many records are in each block. If (R) is the total number of records and (B) is the number of disk blocks, the average number of records per block is (R/B).

Average Access Time: In the worst case, binary search will access ($\log_2(B)$) disk blocks. However, since we are interested in the average access time, we consider that for each level of the binary search, we might access a block. The average number of disk block accesses will be

the sum of disk block accesses across all levels of the search.

The average access time (T) for accessing any record = $\log_2(B) + 1$

The "+1" is for the initial access to the first block.

Thus, the average access time needed to access any record of the given file using binary search is approximately: **$O(\log_2(B))$**

Q64 Text Solution:

The correct answer is **option A**.

To find the total number of students enrolled in each course along with its course name, we need to ensure that we correctly join the STUDENT and COURSE tables and then group by the course name while counting the number of students.

A) SELECT coursename, count(*) 'total'
FROM STUDENT NATURAL JOIN COURSE
GROUP BY coursename;

This query uses a natural join between STUDENT and COURSE. Since courseno is common in both tables, it will join them correctly. The query then groups by coursename and counts the number of students in each course. **This query is correct.**

B) SELECT C.coursename, count(*) 'total'
FROM STUDENT S, COURSE C
WHERE S.courseno = C.courseno
GROUP BY coursename;

This query explicitly joins the STUDENT and COURSE tables using a WHERE clause. It counts the number of students for each course and groups by coursename. **This query is also correct.**

C) SELECT coursename, count(*) 'total'
FROM COURSE C
WHERE courseno IN (SELECT courseno FROM STUDENT);
This query selects coursename from the COURSE table and counts the number of students enrolled in the courses. However, it does not join



the STUDENT table; it only filters the courses that have students enrolled. As a result, it does not count the actual number of students per course, which makes **this query incorrect**.

Q65 Text Solution:

Option- **A(A-III, B -I, C -IV, D - II)**

Explanation:

List-I	List-II	Explanation
A. Degree of Table R	III. 6	The degree of a relation relates to the number of qualities it contains. The relation $R = \text{STUDENT} \times \text{COURSE}$ contains six attributes: Rollno, Name, Courseno (from STUDENT), Courseno (from COURSE), Coursename, and Capacity.
B. Cardinality of Table R	I. 15	Cardinality denotes the number of tuples (rows) in the relation. The Cartesian product yields $N \times M$ tuples, where N is the number of records in STUDENT and M is the number of records in COURSE. So the cardinality is $3 \times 5 = 15$.
C. Foreign key of relation STUDENT	IV. Course no.	Courseno is a foreign key in the STUDENT database that points to the COURSE table's Courseno.
D. Foreign key of relation COURSE	II. NIL	The COURSE table contains no foreign keys that refer to other tables. So the foreign key in COURSE is NIL.

Q66 Text Solution:

Option- **A(57,53)**

Explanation: To find the order of internal and leaf nodes in a B-tree:

- Search key size: 10 bytes
- Block size: 1024 bytes
- Record pointer size: 9 bytes
- Block pointer size: 8 bytes

Step 1: Order of Internal Nodes (K)

- Each internal node entry consists of 10 bytes

(key) and 8 bytes (block pointer), for 18 bytes.

- The number of entries per block for internal nodes is: $1024/18 \approx 57$. Thus, **K=57**

Step 2: Order of Leaf Nodes (L)

- Each leaf node entry consists of 10 bytes (key) and 9 bytes (record pointer), totaling 19.
- The number of entries per block for the leaf node is: $1024/19 \approx 53$. Thus, **L= 53**

Q67 Text Solution:

Option- **C(A and C only)**

Explanation: Given a relation scheme $R(x, y, z, w)$ with functional dependencies $F = \{x \rightarrow y, z \rightarrow w\}$ set . All attributes take single and atomic values only.

1 NF: A relation is 1NF if all attributes have atomic (indivisible) values. Given that all characteristics have single and atomic values, the relation R is 1NF.

Primary key: The primary key is the set of qualities that define all other attributes. Based on the analysis above, the candidate key is : $\{x, z\}$, thus, xz is the primary key of R.

Q68 Text Solution:

Option- **A(Total specialization and overlapping constraints)**

Explanation: Constraints for this scenario:

- Because a part may be manufactured and purchased, it implies overlapping subclasses.
- Every part must be manufactured, purchased, or both, implying total specialization because all parts must be classified into at least one of these subclasses.

Overlapping Constraints: A part can be classified into many subclasses (it can be made and purchased).

Total specialization: Each instance of the parent class (part) needs to be a member of at least one of the subclasses (either bought, made, or both).

Q69 Text Solution:

Option- **B(Either A or C only)**

Explanation: A transaction can shift from ACTIVE to:



FAILED if an error occurs during execution (for example, a hardware failure or a constraint violation that causes the transaction to terminate).

PARTIALLY COMMITTED after completing its execution but before all modifications are permanently applied. At this stage, the transaction is regarded to be in the process of committing, but the changes have not been finalized.

Q70 Text Solution:

Option- **B (DDL)**

Explanation: DDL- Data Definition Language
It is concerned with establishing the schema and database objects and is a subset of SQL (Structured Query Language) that defines, manages, and modifies a database's structure, including its tables, schemas, indexes, views, and constraints.

Q71 Text Solution:

Option- $A(p + \frac{q}{100})$

Explanation: Given,

p: Access time to memory (in nanoseconds).

q: Extra time (in nanoseconds) to handle a page fault.

Page fault rate: For every 100 instructions, there is a one-page fault. $\rightarrow q/100$

Every memory access that doesn't result in a page fault has a normal memory access time (p). Only when a page fault occurs is page fault treatment time (q) added. This occurs once in every 100 accesses because a page failure occurs once for every 100 instructions.

Consequently, the EMAT (effective memory access time) is: $p + \frac{q}{100}$

Q72 Text Solution:

Option- **B (B and C only)**

Explanation:

Indexed allocation: The file system keeps an

index block with pointers to the file's data blocks dispersed throughout the storage in indexed allocation. External fragmentation does not affect indexed allocation because the file blocks may not be contiguous.

Linked Allocation: Each file is represented as a linked list of blocks dispersed over the storage in linked allocation. Blocks do not have to be adjacent, as with indexed allocation, preventing external fragmentation.

Q73 Text Solution:

Option- **B(Both B and C are TRUE only)**

Explanation:

B. Most CPU architectures include a separate interrupt request (IRQ) line or a group of IRQ lines that inform the CPU when an interrupt needs to be handled. This line is used to notify the CPU of an interrupt condition, which causes it to pause its current task and run the interrupt handler.

C. The interrupt vector table is a specific table used by the CPU to store memory addresses for interrupt service routines (ISRs). Whenever an interrupt occurs, the CPU utilizes the interrupt vector to hop to the correct memory address and execute the interrupt handler.

Q74 Text Solution:

Option: **B(A - IV , B - III , C - I , D - II)**

Explanation:

List-I	List-II	Explanation
System Calls	Description	
A. fork()	IV. Creates a child process.	Used to generate new process(child).
B. exec()	III. Loads the specified program in memory.	Replaces the current process image with a new program, essentially putting a new program into memory.



C. kill()	I. sends a signal from one process to particular process	Signals (e.g., terminate, interrupt) can be sent to other processes.
D. exit()	II. Indicates termination of current process	Used to terminate the current process and notify the operating system.

Q75 Text Solution:Option- **C(4.66)**

Explanation: Using Preemptive Shortest Job First scheduling algorithm

Gantt Chart:

Processes	P1	P2	P1	P3
Arrival Time	0-1	1-5	5-12	12-21

Formula:

Turnaround time = Completion time - arrival time

Waiting Time = Turn around time - burst time

Average Waiting Time = Total waiting time / number of processes

P					
r					
o					
c					
e	A	B	C	T	W
s	T	T	T	A	T
s					
s					
e					
s					
P	0	8	1	1	4
1			2	2	
P	1	4	5	4	0
2					
P	2	9	2	1	1
3			1	9	0

Average waiting time = $(4+0+10)/3 = 14/3 = 4.66$ **Q76. Text Solution:**Option- **A(12)**

Explanation: The number of bits reserved for the frame offset corresponds to the size of each frame (which is the same as the page size).

Page size = 4 KB = 2^{12} bytes

The number of bits required to represent the frame offset (the position within the frame) is 12 bits

Q77. Text Solution:Option- **B(2^{20})****Explanation:** Given,

Virtual address space = 32-bit, therefore,

addressable bytes = 2^{32} bytes = 4GBPage size = 4KB = 2^{12} bytes

Calculate the Number of Pages:

$$\text{Number of pages} = \frac{2^{32} \text{ bytes}}{2^{12} \text{ bytes}}$$

$$\text{Number of pages} = 2^{32-12} = 2^{20}$$

Q78. Text Solution:Option- **B(34)**

Explanation: Given, Total physical memory in bytes: 16 GB = $2^4 \times 2^{10} \times 2^{10} \times 2^{10}$ bytes = 2^{34} bytes

Q79. Text Solution:Option- **A(2MB)****Explanation:** Given,

Virtual address space = 32-bit

Page size = 4KB = 2^{12} bytes

Page table entry size = 2 bytes

Number of pages = 2^{20} bytes**Size of page table:**

$$\begin{aligned} \text{Page table size} &= 2^{20} \times 2 \text{ bytes} \\ &= 2^{21} \text{ bytes} = 2 \text{ MB} \end{aligned}$$

Q80. Text Solution:Option- **D(2KB)****Explanation:** Given:

Page size = 4 KB

Process size = 34 KB



$$\text{Pages required} = \left\lceil \frac{34 \text{ KB}}{4 \text{ KB}} \right\rceil = 9$$

Total memory allocated = 36 KB

Internal fragmentation = 36 KB - 34 KB = 2 KB

Q81 Text Solution:

Options: **B(A-II, B - I, C - IV, D - III)**

Explanation:

List-I	List-II	Explanation
A. Odd Function	II. XOR Gate	An Odd Function returns true if an odd number of inputs are true. The XOR gate behaves in this fashion.
B. Universal Gate	I. NAND Gate	The NAND gate is universal.
C. 2421 Code	IV. Self-complementing	Self-complementing codes generate the 9's complement of a decimal number by combining 1's and 0's in the corresponding 2421 form.
D. Buffer	III. Amplification	A buffer is used for amplification, which increases signal strength without affecting the logic state.

Q82 Text Solution:

Option- **A(15.6)₈**

Explanation:

Step 1: Convert Hexadecimal to Binary:

D=1101, C= 1100 (In binary)

The hexadecimal number D.C=1101.1100(In binary)

Step 2: Convert Binary to Octal: Group the binary digits into triplets, starting from the decimal point:

- For the whole number part: 1101 becomes 001 101

- For the fractional part: 1100 becomes 110 000

Now, convert each of these binary triplets to their octal equivalent:

- 001=1, 101=5, 110 =6 and 000=0

So, 1101.1100 in binary is equivalent to 15.60 in octal.

Q83 Text Solution:

Option: **A(16, 8×1)**

Explanation:

1. Number of Multiplexers: With 16 bits in each register and one multiplexer per bit, a total of 16 multiplexers are required.

2. Multiplexer Size: Each multiplexer must select one bit from any of the 8 registers. As a result, each multiplexer is an 8-to-1 multiplexer (8 inputs for the 8 registers and one output for the selected bit).

Thus, 16 multiplexers of size 8×1 are required.

Q84 Text Solution:

Option- **D(HLT)**

Explanation: This is not a pseudo-instruction: HLT (stop). It is a real machine instruction that halts the program's execution. The program is essentially stopped when the HLT instruction is performed since it forces the processor to halt execution.

Q85 Text Solution:

Option- **A(ABCDE+*+*FGH+*/)**

A. Explanation: Conversion Steps:

1. Inner Parentheses:

- (D + E) becomes DE+
- (G + H) becomes GH+

2. Next Operations:

- C * (D + E) becomes CDE+*
- B + C * (D + E) becomes BCDE+*+
- A * (B + C * (D + E)) becomes ABCDE+*+*

3. Second Part:

- F * (G + H) becomes FGH+*

4. Final Division:

- A * { B + C * (D + E) } / { F * (G + H) } becomes ABCDE+*+*FGH+*/

Q86 Text Solution:

Option- **D(A, D, C, B)**

Explanation: The increasing order of complexity



is: A. I/O Module < D. DMA < C. I/O Channel < B. I/O Processor.

Options	Explanation
A. I/O Module	One piece of hardware that controls communication between the computer's central processing unit and external peripherals is the I/O module.
B. DMA	More complicated than an I/O module, but it still functions somewhat simply and solely controls memory access for I/O activities.
C. I/O Channel	By controlling several I/O devices and taking care of things like starting I/O operations and buffering, it offers a greater degree of control over I/O activities than an I/O module or DMA.
D. I/O Processor	The specialized processor that relieves the CPU of I/O duties, such as controlling data transfers, carrying out I/O orders, and carrying out I/O-related calculations.

Q87 Text Solution:

Option- C(A and C only)

Explanation: A Combinational circuit is a kind of digital circuit that has no memory of previous states and whose output is only controlled by its current inputs. It has no memory components or feedback loops, and as inputs change, the output instantly adapts.

A. Combinational circuit has no memory of past states; instead, the output is immediately dictated by the present inputs.

C. There are no feedback channels in combinational circuits. This indicates that the inputs are not influenced by the outputs.

Q88 Text Solution:

Option B is Correct

(A) Multiport Memory:

Multiport memory facilitates multiple processors accessing the same memory simultaneously. However, it does not directly address cache coherence, as it does not ensure that the cached

copies remain consistent across processors.

Hence, Statement A is Incorrect.

(B) Allow Only Nonshared Data to Be Stored in Cache:

By caching only nonshared data, the cache coherence problem is avoided entirely. However, this is not a scalable solution in systems where shared data is common, but it is technically a valid solution. Hence, Statement B is Correct.

(C) Using a Snoopy Cache Controller:

A snoopy cache controller uses coherence protocols (e.g., MESI) to monitor and ensure consistency of cached data by broadcasting updates or invalidations over the shared bus. This is an efficient and widely-used solution for the cache coherence problem. Hence, Statement C is Correct.

(D) Using Memory Interleaving:

Memory interleaving improves memory access speed by splitting memory into banks for parallel access. It does not address cache coherence, as it focuses on access optimization rather than consistency. Hence, Statement D is Incorrect.

Q89 Text Solution:

Option A is Correct

Statement I: CISC computers have a large number of addressing modes.

This Statement is True as CISC (Complex Instruction Set Computing) architectures are designed to handle complex instructions in a single instruction, which often requires multiple addressing modes to access operands efficiently. Examples include immediate, direct, indirect, indexed, etc.

Statement II: In RISC machines, memory access is limited to load and store instructions.

This Statement is also True as RISC (Reduced Instruction Set Computing) architectures are designed with a simplified instruction set, where memory operations are restricted to explicit load and store instructions. This helps streamline instruction execution and improves performance.

Q90 Text Solution:

Option B is Correct



A. Control memory is part of the hardwired control unit.

This Statement is False as Control memory is a feature of the microprogrammed control unit, not the hardwired control unit. In a microprogrammed control unit, control signals are stored in control memory, while a hardwired control unit uses combinational circuits.

B. Program control instructions are used to alter the sequential flow of the program.

This Statement is True as Program control instructions (e.g., jump, branch, call, return) are explicitly designed to modify the sequential execution of instructions by altering the program counter (PC).

C. The register indirect addressing mode for accessing memory operand is similar to displacement addressing mode.

This Statement is False as the Register indirect addressing mode accesses the memory address stored in a register directly, while displacement addressing mode adds an offset (displacement) to the value in the register to calculate the effective address.

D. CPU utilization is not affected by the introduction of Interrupts.

This Statement is False as Interrupts improve CPU utilization by allowing the CPU to handle I/O or other tasks asynchronously instead of waiting. However, they do have an impact on the overall CPU operation.

Q91 Text Solution:

Option A is Correct

List I	List II	Explanation
A. Data Link Layer	IV. Piggybacking	The Data Link Layer combines acknowledgment with data transmission using piggybacking to optimize performance.

List I	List II	Explanation
B. Network Layer	III. Productive billing information	The Network Layer provides routing and forwarding, and is responsible for producing network-related statistics like billing.
C. Transport Layer	I. True end-to-end layer	The Transport Layer ensures reliable data delivery between source and destination, acting as the true end-to-end layer.
D. Session Layer	II. Token Management	The Session Layer manages dialogs between devices and controls the flow using token management mechanisms.

Q92 Text Solution:

Option A is Correct

Telnet, FTP, and HTTP are Application Layer protocols in the OSI and TCP/IP models. These protocols operate at the application layer and provide services for end-users:

- Telnet: Remote login protocol.
- FTP: File Transfer Protocol for file sharing.
- HTTP: HyperText Transfer Protocol used for web communication.

Hence, This Statement is True

Statement II:

The Iridium Project involved launching 66 low-earth orbit satellites to create a global communication system. The project was successfully launched in the 1990s by Iridium Communications Inc., aiming to provide voice and data communication globally.

Hence, This Statement is also True

Q93 Text Solution:

Option A: This Statement is True as Packet switching dynamically allocates bandwidth, leading to better utilization of resources compared to circuit switching, which reserves a fixed path and bandwidth even if idle.

Option B: This Statement is False as Packet switching typically results in more variation in delay (jitter) than circuit switching. This is



because packets may take different routes to reach the destination, leading to delay variability. In contrast, circuit switching provides a dedicated path, ensuring consistent delay.

Option C This Statement is True as In packet switching, the sender and receiver can use different bit rates, formats, or framing methods, as packets can be reassembled at the destination. Circuit switching requires fixed and synchronized communication.

Option D This Statement is True as Packet switching may lead to packet reordering because packets can take different paths in the network. Circuit switching ensures packets are received in order as they travel a dedicated path.

Q94 Text Solution:

Option A is Correct

In Manchester Encoding, A single bit is encoded into two signal elements: A transition in the middle of the bit duration (High-to-Low or Low-to-High), and This ensures synchronization and clock recovery.

Bit Rate is Number of bits transmitted per second. And, Baud Rate is Number of signal changes (symbols) per second. In Manchester encoding, each bit is encoded as two signal elements, making the baud rate twice the bit rate. Thus, the bit rate is half the baud rate.

Q95 Text Solution:

Option D is Correct

In Class B Addresses, Default subnet mask: 255.255.0.0 (16 bits for the network ID, 16 bits for the host ID). Now, Subnetting with a 6-bit Subnet Number:

- 6 bits are borrowed from the host bits to create subnets.
- Total subnet mask becomes 255.255.252.0 or /22 (16 network bits + 6 subnet bits).

So, the Maximum Number of Subnets:

With 6 bits for subnetting: $2^6=64$ subnets.

However, 2 subnets are reserved (one for the all-zeros subnet and one for the all-ones subnet).

Thus: Usable subnets= $64-2=62$ subnets.

Total host bits = $16 - 6 = 10$ host bits (after borrowing 6 bits for subnetting).

So, the Number of hosts per subnet are $2^{10}-2$
= 1022 hosts

(subtract 2 for network and broadcast addresses).

Q96 Text Solution:

Option D is Correct

MIME (Multipurpose Internet Mail Extensions) is an extension of the basic email protocol that allows the transmission of multimedia content, such as images, audio, video, and non-ASCII text (e.g., attachments). MIME does not replace existing protocols like SMTP but works with them to encode and decode multimedia data. MIME extends the capability of email systems to support non-textual data and multiple types of attachments. Without MIME, emails are limited to plain text.

Whereas, IMAP (Internet Message Access Protocol) is Used to retrieve and manage emails stored on a mail server. It does not handle multimedia.

SMTP (Simple Mail Transfer Protocol) Handles the sending of plain text email but does not inherently support multimedia. MIME is used alongside SMTP for multimedia support.

POP3 (Post Office Protocol 3) is Used to retrieve emails from a server, but it does not include multimedia handling capabilities.

Q97 Text Solution:

Option A is Correct

To ensure secure communication, the receiver's public key is used by the sender for encryption, making it impossible for others (without the private key) to decrypt the message.

Q98 Text Solution:

Option B is Correct

A. Frequency division multiplexing (FDM) can be handled by digital circuits:

This Statement is False as FDM is primarily an analog multiplexing technique where multiple



signals are transmitted on different frequency bands. It does not require digital circuits.

B. Time division multiplexing (TDM) can be handled by analog circuits:

This Statement is False as TDM is a digital multiplexing technique, where time slots are allocated to each signal. It cannot be handled by analog circuits.

C. Wavelength division multiplexing (WDM) is used with optical fiber for combining two signals:

This Statement is True as WDM is specifically used in optical fiber communication to combine multiple optical signals of different wavelengths into a single optical fiber.

D. Frequency division multiplexing (FDM) can be applied when the bandwidth of a link is greater than the bandwidth of the signals to be transmitted:

This Statement is True as FDM requires a bandwidth greater than the sum of all signals' bandwidths to allocate non-overlapping frequency bands to each signal.

Q99 Text Solution:

Option A is Correct

A. X.25 is connection-oriented network:

This Statement is True as X.25 is a connection-oriented protocol suite designed for packet-switched networks. It establishes a virtual circuit before transmitting data, ensuring reliability.

B. X.25 doesn't support switched virtual circuits:

This Statement is False as X.25 does support switched virtual circuits (SVCs), which are dynamically established and terminated on

demand. It also supports permanent virtual circuits (PVCs).

C. Frame relay service provides acknowledgments:

This Statement is False as Frame relay is a best-effort service and does not provide end-to-end acknowledgments or flow control. It relies on higher-layer protocols (e.g., TCP) for reliability.

D. Frame relay service provides detection of transmission errors:

This Statement is True as Frame relay includes a Cyclic Redundancy Check (CRC) in its frames to detect transmission errors. However, it does not correct these errors—it simply drops erroneous frames.

Q100 Text Solution:

Option A is Correct

Cyclic Redundancy Check (CRC) detects transmission errors by using a generator polynomial $G(x)$. It ensures certain properties to detect specific types of errors. A polynomial $G(x)$ can detect all errors with an odd number of inverted bits if and only if $(x+1)$ is a factor of $G(x)$.

Why $(x+1)$ Detects Odd Bit Errors?

$(x+1)$ corresponds to binary 11, which ensures that the CRC calculation detects all odd-parity errors. When a generator polynomial includes $(x+1)$ as a factor, it guarantees that the resulting checksum is sensitive to any error with an odd number of bit flips, as such errors will always generate a non-zero remainder.

